FYP-2

Final Evaluation Report

F22-118-D

LearnLive

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Date: 4th June, 2023

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Chapter 1: Introduction

LearnLive is an online platform, with all the operations of having an online coding class, with live access to teachers for student's code on their screen, chat for parents and teachers for uninterrupted and transparent communication, and LMS for sharing materials and developing lesson plans. It is an industry-funded project by **WEEPRO CODERS**.

Chapter 2: Project Vision

2.1 Problem Statement

The problem for WEEPRO CODERS is that they want to monitor and track their coding class along with uninterrupted communication while relying on multiple applications and limited human resources, for example;

1. Zoom is being used for online classes with another tool (Scratch/Replit) for programming included in the session, in case of interactive quizzes, they include Kahoot as another app in the zoom call by logging the results and sessions separately
2. Students can't share their WEEPRO CODERS's Projects on social media or to their friends
3. Parents communicate on WhatsApp and they also get the weekly challenges and necessary announcements in the groups which have a chance of missing as well.
4. This all together builds a lot of communication for WEEPRO CODERS executive team to track or monitor or participate vigilantly.
5. Communication with teachers and supporting roles on separate WhatsApp groups regarding execution or feedback on their classes and also if there are any specific concerns from the parents
6. Materials are being shared using Google Drive, learners will be working independently so we can't access or monitor the progress until work is not submitted, Evaluation is done manually to generate the Leader board
7. To strategize for further coding camps, they generate data trends using python scripts on data from various excel sheets

2.2 Motivation

Our main motivation is to make this whole process of online teaching efficient as well as convenient by providing a conducive learning environment by making a platform that integrates the features of WhatsApp, Zoom, Replit (A collaborative browser-based IDE), Slack and Google Classroom on a single platform. It will have three types of users: Teachers, Students and Parents.

2.3 Objective

Our web based platform will have three main stakeholders: Teachers, Students and Parents. The objectives of these stakeholders will be as following:

- Teachers will have an analytics dashboard that will give them vital information for analysis purpose for example: Number of students, their retention rate, how many people are enrolling, from which cities people are enrolling, which time slots suit students the best, which teacher has the most classes, which age group to target, future trends etc. This would involve a lot of learning and visualization of data. Teachers will be able to communicate with students as well as their parents through the application, give online classes to students, enable students to ask any queries during off time like a forum where even other

students can help them, help students code during class and see any issues they have through an in-built IDE in the web application and give homework/quizzes/tests to students on the application.

- Parents will be able to directly communicate with the teachers and see different information regarding their children e.g., how they are performing in the programme as well as their attendance etc.
- Students will be able to take online classes, ask any queries just like slack, code on the platform, perform tasks given by the teacher like HomeWorks and quizzes and communicate directly with the teacher.
- We will also develop a chatbot for the web application in case any of the users have any FAQs or find it difficult to use the website. It is especially important to make it for the parents who are generally not tech savvy.

2.4 Outcome

The final outcome of this whole project would be a web-based platform that is very easy to use and making the whole process of online teaching convenient for all the parties involved.

2.5 Target Audience

Target audience is mainly WEEPRO Coders instructors, students and parents who will be using this platform.

Chapter 3: Requirements and Design

This section covers the basis of the requirements and design related to LearnLive.

3.1 User Classes and Characteristics

Standard users may belong to any demographic group including any gender, nationality that can use a computer's browser. The site requires its users to be above 6 years of age. It does require its users to have any specific computer knowledge. User interface is in the English language so users should have basic English reading knowledge.

Stakeholders	**Interests**	**Constraints**
Standard User	Using this platform to perform certain activities like Student can upload his/her assignments	None
Developer	Building of the actual algorithm and structure of the system	None
Advertiser	Sponsored advertisements of products on the platform	None
Product Owner	Supervision of the team of developers, testers and designers	None
Designer	The UI/UX design of the system	None
Tester	Testing of the system for potential bugs	None

3.2 Operating Environment

3.2.1 Hardware Requirements

- Pentium-IV (Processor).
- 512 MB Ram
- 512 KB Cache Memory
- Hard disk 10 GB

- Microsoft/Apple Compatible keyboard.

3.2.2 Software Requirements

- Operating System : Windows/MacOS
- Web browser: Google Chrome, Mozilla Firefox for Windows or MacOS.
- Programming language: Javascript.
- Web-Technology able to run: Node.JS, REACT.js.
- Back-End: Node.js server, Express.js, MongoDB server.
- Web Server: AWS

3.2.3 Design and Implementation Constraints

1. The website will support the following browsers: Microsoft Edge, Chrome 27.0+ for both Windows and MacOS and Safari for MacOS.
2. The website supports English language only.
3. The website requires certificates issued by CA to use HTTPS.
4. The website is built on Node Js and React Js libraries.
5. The website uses “jsonwebtoken” library for generating random tokens every time a user logins in the website

3.2.4 User Documentation

The system will not provide any documentation. Online written and video tutorials will be served when users register to the site.

3.2.5 Assumptions and Dependencies

1. There will be availability of Internet via 3G, 4G, 5G or Wi-Fi.
2. The user of the site should be acquainted with the English language.
3. The user should have a valid email address in order to register into the website.
4. Central server of the system must be able to handle all the incoming requests simultaneously.
5. The user has basic knowledge of computers and the Internet.
6. The user has the understanding of cyber laws and crimes as well as basic societal ethics and should respect the privacy of others.

3.3 High Level Features

- Online coding class with video call and collaborative coding.
- Chatbot to answer frequently asked queries of parents.
- Three-way communication between Teachers, Students and Parents.
- Monitoring Student attendance and active time in the class

- Conducting interactive quizzes and logging the performance
- Workbench for Teachers; Designing Lesson Plans using Templates and utilizing previous resources, Designing and Assigning Weekly Challenges with Checkpoints and score scheme, Leaders board (shareable) through evaluation.
- Analytics Dashboard.

3.4 Functional Requirements

This section describes the functional requirements of each user of LearnLive i.e. Admin, Teacher, Student and Parent.

3.4.1 Functional Requirements for Admins

Admin can:

- Log in using valid credentials
- Schedule Classes (no conflicts) and also this will become the timetable to be displayed weekly or monthly (calendar)
- Assigning Learners and Parents to the camp
- Assigning Teachers to the camp
- Ability to set permissions for each profile
- Attendance log and active hours of each participant
- Edit profile of themselves as well as other users who are not admins

3.4.2 Functional Requirements for Learners

Learners can:

- Attend the classes
- Participate in weekly challenges or projects
- Perform quizzes
- Submit assignments
- Access to the leaderboard and star of the week
- Download digital certificate
- Code online with in-built compiler
- Submit code for review
- Send messages to the teacher
- Post queries
- Edit their profile

3.4.3 Functional Requirements for Parents

Parents can:

- Send messages to the teacher
- View certificate

- View class task, weekly challenge and project being set for the learners
- Give feedback on class tasks
- View leaderboard
- See their child's progress through dashboard analytics
- View attendance of their child
- Edit their profile

3.4.4 Functional Requirements for Teachers

Teachers can:

- Schedule classes
- Upload assignments
- Prepare interactive quizzes
- Review codes
- Grade assignments and projects
- Take attendance
- Generate certificates for learners
- Edit the leaderboard information
- Message student and parent
- Edit their profile

3.5 Non-Functional Requirements

The non-functional requirements of the application are:

- System can be accessed by millions of users simultaneously
- System will provide Informative Feedback
- System will permit easy reversal of actions
- System will be available 24/7 as reliable cloud servers will be used
- System application will be compatible with all browsers
- System will respond to the user requests in a quick manner
- System's user interface will be easy to use and learn
- System will deal with exception handling
- System will allow users to apply previously learned knowledge to new tasks
- System will be easy to navigate with the help of hyperlinks thus reducing short term memory load
- The system will be easily maintained because it will be built in modules and if a module is to be changed the whole system doesn't change. The backend model can also be updated and replaced easily

Chapter 4: Timeline

4.1 Learning Management System

Date: ***September 2022 - October 2022***

Teacher Panel:

- Designing and providing class tasks, Weekly Challenges and Projects with Checkpoints and score scheme
- Preparing interactive quizzes
- Generate certificates at the end of the coding camp.

Learner Panel:

- Developing weekly challenges or projects uploaded by teachers.
- Performing quizzes

4.2 Live coding classes

Date: ***November 2022 - December 2022***

Admin Panel:

- Schedule Classes and also this will become the timetable to be displayed weekly or monthly (calendar)
- Assigning learners to the camp
- Assigning Teachers to the Camps
- Control Panel for setting the permissions for all
- Notifications about each class being started, teacher and student joined
- Class starting, ending time and attendance log and active hours of each participant

Learner Panel:

- They can join a scheduled class, early reminders and notifications for joining late included
- Performing quizzes
- Performing coding tasks

Teacher panel:

- Joining Scheduled class, early reminders and notifications for joining late included

4.3 Chatting Channel/ Discussion Forums

Date: ***January 2023 – February 2023***

- Chat between teachers and students
- Chat between teachers and parents

4.4 Collaborative Coding Platform

Date: ***February 2023 - March 2023***

- Coding Class with the provision of 9 students in a class and one teacher. Summing it all make 10 users in a class with varying controls and rights
- Access of teacher and team to student's code

- Control panel for teachers to allow access to students' for peer review or peer task

4.5 Learner Assessment

Date: ***April 2023 - May 2023***

Teacher panel:

- Evaluating weekly challenges, projects and quizzes.
- Preparing Leaders board through evaluation.
- Code Reviews
- Authenticating Certificates to the learners of the camp

Admin Panel:

- Generate certificates at the end of the coding camp.
- Tracking of students' attendance and active time

Parent Panel:

- View Leaders board and access to the detailed evaluation and feedback to identify the gaps
- View Certificate
- Share their Parent review as a video or text message and share rating for the camp, teacher and team

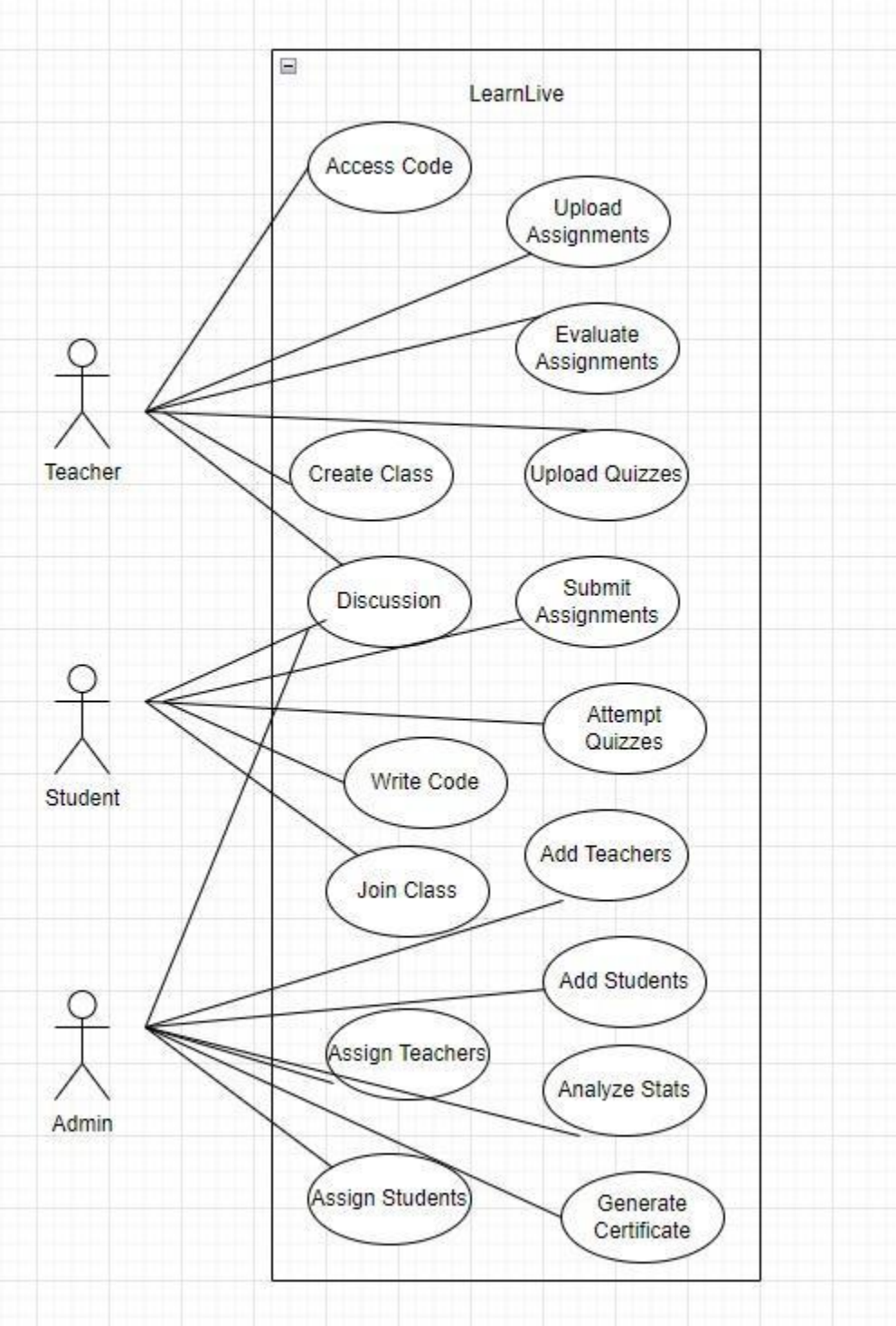
4.6 Analytics Dashboard

Date: ***May 2023- June 2023***

- Current number of students
- Students' retention rate
- How many people are enrolling?
- From which cities people are enrolling?
- Which time slots suit students the best?
- Which teacher has the most classes?

Chapter 5: Use Cases

5.1 Use Case Diagram



5.2 High Level Use Cases

5.2.1 Use Case 1

Use case ID	UC-1
Use case:	Login
Primary Actor	Admin, Student, Teacher and Parent
Type	Primary
Description	Users input their credentials on the login page. The system then verifies the user and redirects to the home screen.

5.2.2 Use Case 2

Use case ID	UC-2
Use case:	Logout
Primary Actor	Admin, Student, Teacher and Parent
Type	Primary
Description	The user will press the logout button and will be redirected to the login screen

5.2.3 Use Case 3

Use case ID	UC-3
Use case:	Upload Assignments
Primary Actor	Teacher
Type	Primary
Description	Teacher can upload assignments under the assignment tab

5.2.4 Use Case 4

Use case ID	UC-4
Use case:	Upload Quizzes
Primary Actor	Teacher
Type	Primary
Description	Teacher can upload quizzes under the quizzes tab

5.2.5 Use Case 5

Use case ID	UC-5
Use case:	Submit Assignments
Primary Actor	Student
Type	Primary
Description	Student can submit the given assignment under the assignment tab

5.2.6 Use Case 6

Use case ID	UC-6
Use case:	Submit Quizzes
Primary Actor	Student
Type	Primary
Description	Student can submit the given quizzes under quizzes tab

5.2.7 Use Case 7

Use case ID	UC-7
Use case:	Grade Quizzes
Primary Actor	Teacher
Type	Primary
Description	Teacher can grade quizzes submitted by students

5.2.8 Use Case 8

Use case ID	UC-8
Use case:	Join/Take Class
Primary Actor	Student
Type	Primary
Description	Student can join/take a class made by the teacher

5.2.9 Use Case 9

Use case ID	UC-8
Use case:	View classes
Primary Actor	Student
Type	Primary
Description	Student can view the classes made by the teacher

5.2.10 Use Case 10

Use case ID	UC-10
Use case:	Create a class
Primary Actor	Teacher

Type	Primary
Description	Teacher can create a zoom class

5.2.11 Use Case 11

Use case ID	UC-11
Use case:	Add/assign teachers
Primary Actor	Admin
Type	Primary
Description	Admin will add or assign teachers to a camp

5.2.12 Use Case 12

Use case ID	UC-12
Use case:	Add/assign students
Primary Actor	Admin
Type	Primary
Description	Admin will add or assign students to camp

5.2.13 Use Case 13

Use case ID	UC-13
Use case:	Analyze Stats
Primary Actor	Admin
Type	Primary
Description	Admin will be able to analyze stats of different things

5.2.14 Use Case 14

Use case ID	UC-14
Use case:	Generate Certificate
Primary Actor	Admin
Type	Primary
Description	Admin will generate certificate at the end of camp completion

5.2.15 Use Case 15

Use case ID	UC-15
Use case:	Discussion
Primary Actor	Student/Teacher
Type	Primary
Description	Student and teacher will be able to communicate with each other

5.2.16 Use Case 16

Use case ID	UC-16
Use case:	Write Code
Primary Actor	Student
Type	Primary
Description	Student will be able to code and get the output

5.2.17 Use Case 17

Use case ID	UC-17
Use case:	Collaborate Code
Primary Actor	Student/Teacher

Type	Primary
Description	Student and teacher will be able to collaborate code with each other

5.3 Extended Use Cases

5.3.1 Use Case: Login

Use Case ID	UC-1
Use Case Name	Login
Actor	Admin, Student, Teacher and Parent
Summary	The user will press the logout button and will be redirected to the login screen.
Pre-Conditions	The user has already signed up
Post-Conditions	The user logins successfully and a prompt of "Login Success" will be displayed
Special Requirements	None
Normal Path	**System Response**
1. User opens the login screen 3. User inputs his registered email and password	2. The login screen is displayed requiring email and password as inputs 4. The home screen opens after verification of the user credentials
Alternative Path	**System Response**
3. User inputs invalid login credentials (either email or password)	4-A. Error message will be displayed: *Invalid Login Credentials*
Frequency of Use	Low

5.3.2 Use Case: Logout

Use Case ID	UC-2
Use Case Name	Logout
Actor	Admin, Student, Teacher and Parent
Summary	Users input their credentials on the login page. The system then verifies the user and redirects to the home page.
Pre-Conditions	User must be already logged in
Post-Conditions	The user successfully logs out of the application and login screen appears
Special Requirements	None
Normal Path	**System Response**
1. User presses the logout button	2. The User logs and login screen appears
No Alternative Path	**No System Response**
Frequency of Use	Low

5.3.3 Use Case: Upload Assignments

Use Case ID	UC-3
Use Case Name	Upload Assignments
Actor	Teacher
Summary	Teacher can upload assignments under the assignment tab
Pre-Conditions	Teacher has already been logged in and is able to see assignments tab

Post-Conditions	Teacher successfully uploads the assignment under the assignment tab and students can view assignments
Special Requirements	None
Normal Path	**System Response**
1. Teacher opens the assignment tab 3. Teacher clicks on "Upload Assignment" button and uploads the assignment	2. "Upload Assignment" button is displayed on the screen 4. The selected assignment is displayed on the screen
Alternative Path	**System Response**
3. Teacher gets the error on clicking "Upload Assignment" button	4-A. Error message will be displayed: *Please select a pdf or word file*
Frequency of Use	Medium

5.3.4 Use Case: Upload Quizzes

Use Case ID	UC-4
Use Case Name	Upload Quizzes
Actor	Teacher
Summary	Teacher can upload quizzes under the Quiz Tab
Pre-Conditions	Teacher has already been logged in and is able to see quizzes tab
Post-Conditions	Teacher successfully uploads the quizzes and students can view them
Special Requirements	None
Normal Path	**System Response**
1. Teacher opens the quizzes tab 3. Teacher clicks on "Upload Quiz" button and uploads the quiz	2. "Upload Quiz" button is displayed on the screen 4. The selected quiz is displayed on the screen

Alternative Path	**System Response**
3. Teacher gets the error on clicking “Upload Quiz” button	4-A. Error message will be displayed: *Please select a link*
Frequency of Use	Medium

5.3.5 Use Case: Submit Assignments

Use Case ID	UC-5
Use Case Name	Submit Assignment
Actor	Student
Summary	Student can submit assignments under the assignment tab
Pre-Conditions	Student has already been logged in and is able to see assignments tab
Post-Conditions	Student successfully uploads the assignment under the assignment tab and teacher can view assignments
Special Requirements	None
Normal Path	**System Response**
1. Student opens the assignment tab 3. Student clicks on “Upload Assignment” button and uploads the assignment	2. “Upload Assignment” button is displayed on the screen 4. The selected assignment is displayed on the screen
Alternative Path	**System Response**
3.Student gets the error on clicking “Submit Assignment” button	4-A. Error message will be displayed: *Please select a pdf or word file*
Frequency of Use	Medium

5.3.6 Use Case: Submit Quizzes

Use Case ID	UC-6
Use Case Name	Submit Quizzes
Actor	Student
Summary	Student can upload quizzes under the Quiz Tab uploaded by the teacher
Pre-Conditions	Student has already been logged in and is able to see quizzes tab
Post-Conditions	Student successfully uploads the quizzes and teacher can view them
Special Requirements	None
Normal Path	**System Response**
1. Student opens the quizzes tab 3. Student clicks on "Upload Quiz" button and uploads the quiz	2. "Upload Quiz" button is displayed on the screen 4. The selected quiz is displayed on the screen
Alternative Path	**System Response**
3. Student gets the error on clicking "Submit Quiz" button	4-A. Error message will be displayed: *Please select a link*
Frequency of Use	Medium

5.3.7 Use Case: Grade Quizzes

Use Case ID	UC-7
Use Case Name	Grade Quizzes
Actor	Teacher
Summary	Quizzes uploaded by teacher and submitted can be grade automatically
Pre-Conditions	Quiz has been uploaded and submitted by teacher and student respectively

Post-Conditions	Quiz will be graded
Special Requirements	None
Normal Path	**System Response**
1. Student will submit quiz 3. Student can view marks	2. Will grade automatically
Alternative Path	**System Response**
1. Student couldn't submit quiz due to some error	2-A. Was not able to receive quiz
Frequency of Use	Medium

5.3.8 Use Case: Join/Take a Class

Use Case ID	UC-8
Use Case Name	Join/take a class
Actor	Student
Summary	Student is able to join/take a zoom class created by the teacher
Pre-Conditions	The class has been created by the teacher
Post-Conditions	Student would be among attendants in the class
Special Requirements	None
Normal Path	**System Response**
1. View classes 2. Click on join desired class	3. Class joined successfully
Alternative Path	**System Response**
1. Student couldn't join meeting due to invalid zoom ID	2. Zoom ID Invalid
Frequency of Use	Medium

5.3.9 Use Case: View classes

Use Case ID	UC-9
Use Case Name	View classes
Actor	Student
Summary	Student is able to view all the classes that are active and been created by a teacher
Pre-Conditions	The classes have been created by the teacher
Post-Conditions	Would be able to see all classes
Special Requirements	None
Normal Path	**System Response**
1. View classes	2. List of classes
Alternative Path	**System Response**
1. No classes visible	2. Teacher has not created any classes
Frequency of Use	Medium

5.3.10 Use Case: Create class

Use Case ID	UC-10
Use Case Name	Create class
Actor	Teacher
Summary	Teacher is able to create a zoom class
Pre-Conditions	Teacher is logged into the website
Post-Conditions	Class is created and students can join in
Special Requirements	None
Normal Path	**System Response**

1. Enter meeting/class name 2. Click on create	3. Class has been created successfully
Alternative Path	**System Response**
1. Invalid zoom ID	2. Teacher needs to login website again
Frequency of Use	Medium

5.3.11 Use Case: Add/assign teachers

Use Case ID	UC-11
Use Case Name	Add/assign teachers
Actor	Admin
Summary	Admin can add or assign teachers in the camp
Pre-Conditions	Admin has already been logged in and able to see add teachers tab
Post-Conditions	Admin successfully assign teachers to the camp
Special Requirements	None
Normal Path	**System Response**
1. Admin opens the add teachers tab 3. Admin fill the form and clicks on "Add Teacher" button and adds the teacher	2. Form and a "Add Teacher" button is displayed on the screen 4. Teacher is added to the camp
Alternative Path	**System Response**
3. Admin gets the error on clicking "Add Teacher" button	4-A. Error message will be displayed: *Please fill the form correctly*
Frequency of Use	Medium

5.3.12 Use Case: Add/assign students

Use Case ID	UC-12
Use Case Name	Add/assign students
Actor	Admin
Summary	Admin can add or assign students in the camp
Pre-Conditions	Admin has already been logged in and able to see add students tab
Post-Conditions	Admin successfully assign students to the respective camp
Special Requirements	None
Normal Path	**System Response**
1. Admin opens the add students tab 3. Admin fill the form and clicks on "Add Student" button and adds the student	2. Form and a "Add Student" button is displayed on the screen 4. Student is added to the camp
Alternative Path	**System Response**
3. Admin gets the error on clicking "Add Student" button	4-A. Error message will be displayed: *Please fill the form correctly*
Frequency of Use	Medium

5.3.13 Use Case: Analyze Stats

Use Case ID	UC-13
Use Case Name	Analyze Stats
Actor	Admin
Summary	Admin will be able to analyze stats of different things
Pre-Conditions	Admin has already been logged in and able to see view analytics
Post-Conditions	Admin successfully analyze stats

Special Requirements	None
Normal Path	**System Response**
1. Admin opens the View Analytics Tab 3. Admin views the different charts/numbers successfully	2. Different charts/numbers are displayed on screen 4. Stats have been successfully analyzed
Alternative Path	**System Response**
3. Admin gets the error on viewing charts/numbers	4-A. Error message will be displayed: *Error getting information*
Frequency of Use	Medium

5.3.14 Use Case: Generate Certificate

Use Case ID	UC-14
Use Case Name	Generate Certificate
Actor	Admin
Summary	Admin is able to generate certificate at the camp completion
Pre-Conditions	Admin has already been logged in and able to see generate certificate
Post-Conditions	Admin successfully generates a certificate for the student
Special Requirements	None
Normal Path	**System Response**
1. Admin opens the Generate Certificate of a student 3. Admin selects the issue date and clicks on Generate Certificate button	2. All relevant details of that student is displayed 4. A certificate is generated on Student's side
Alternative Path	**System Response**

3. Admin gets the error on viewing certificate details	4-A. Error message will be displayed: *Error getting details*
Frequency of Use	Medium

5.3.15 Use Case: Discussion

Use Case ID	UC-15
Use Case Name	Discussion
Actor	Student/Teacher
Summary	Student and Teacher will be able to communicate with each other
Pre-Conditions	Student and teacher are already logged in and are able to see Discussion Tab
Post-Conditions	Student and teacher are able to chat successfully
Special Requirements	None
Normal Path	**System Response**
1. Student and Teacher are able to see Discussion Tab 3. Student and Teacher send messages to communicate with each other	2. Discussion forums can be seen. 4. Messages are delivered successfully and they are able to communicate successfully
Alternative Path	**System Response**
3. Student and Teacher gets the error on viewing Discussion Forum	4-A. Error message will be displayed: *Error getting certain details*
Frequency of Use	Medium

5.3.16 Use Case: Write Code

Use Case ID	UC-16
Use Case Name	Write Code

Actor	Student
Summary	Student will be able to code and get output
Pre-Conditions	Student has already been logged in and able to see code editor
Post-Conditions	Student successfully codes and gets the output
Special Requirements	None
Normal Path	**System Response**
1. Student opens the Code Editor Tab 3. Student writes the code	2. Code editor can be seen 4. Code gets compiled and returns the output
Alternative Path	**System Response**
3. Admin gets the error on viewing code editor	4-A. Error message will be displayed: *Error loading code editor*
Frequency of Use	Medium

5.3.17 Use Case: Collaborate Code

Use Case ID	UC-17
Use Case Name	Collaborate Code
Actor	Student/Teacher
Summary	Student and Teacher will be able to collaborate code with each other
Pre-Conditions	Student and teacher have already been logged in and able to see collaborate code
Post-Conditions	Student and teacher are able to collaborate code successfully
Special Requirements	None
Normal Path	**System Response**
1. Student and Teacher opens the Code Editor	2. Code Editor is displayed on the screen

and generate the room code 3. Student and teacher exchange the room code through discussion forum	4. Student or Teacher enters the room code and are able to join for collaborative coding
Alternative Path	**System Response**
3. Student or Teachers gets the error on opening room	4-A. Error message will be displayed: *Error getting room ID*
Frequency of Use	Medium

5.4 Operation Contracts

5.4.1 Contract Use Case : UC-5

Operation:	Submit Assignment
Cross References:	Upload Assignment
Pre-Conditions: • The assignment has been uploaded by the teacher • Student has submitted valid file format • The file has been submitted before the deadline	
Post Conditions: • Teacher can view the submitted file • Teacher has marked the assignment	

5.4.2 Contract Use Case : UC-6

Operation:	Submit Quiz
Cross References:	Upload Quiz
Pre-Conditions: • The quiz has been uploaded by the teacher • Student has submitted the quiz before deadline • Student has completed the quiz before the timer ran out	

Post Conditions: • System has marked the quiz automatically • System has displayed the students marks

5.4.3 Contract Use Case : UC-7

Operation:	Grade Quiz
Cross References:	Upload Quiz and Submit Quiz
Pre-Conditions: • The quiz has been uploaded by teacher • The quiz has been submitted by student	
Post Conditions: • Quiz has been checked • System updates the marks automatically • Student can view marks	

5.4.4 Contract Use Case : UC-8

Operation:	Join/Take Class
Cross References:	Create Class and View Classes
Pre-Conditions: • The zoom class has been created by teacher • Student can view the zoom class link uploaded by teacher	
Post Conditions: • Student clicks the link • Student is among the attendants in the class • Student can take the class	

5.4.5 Contract Use Case : UC-14

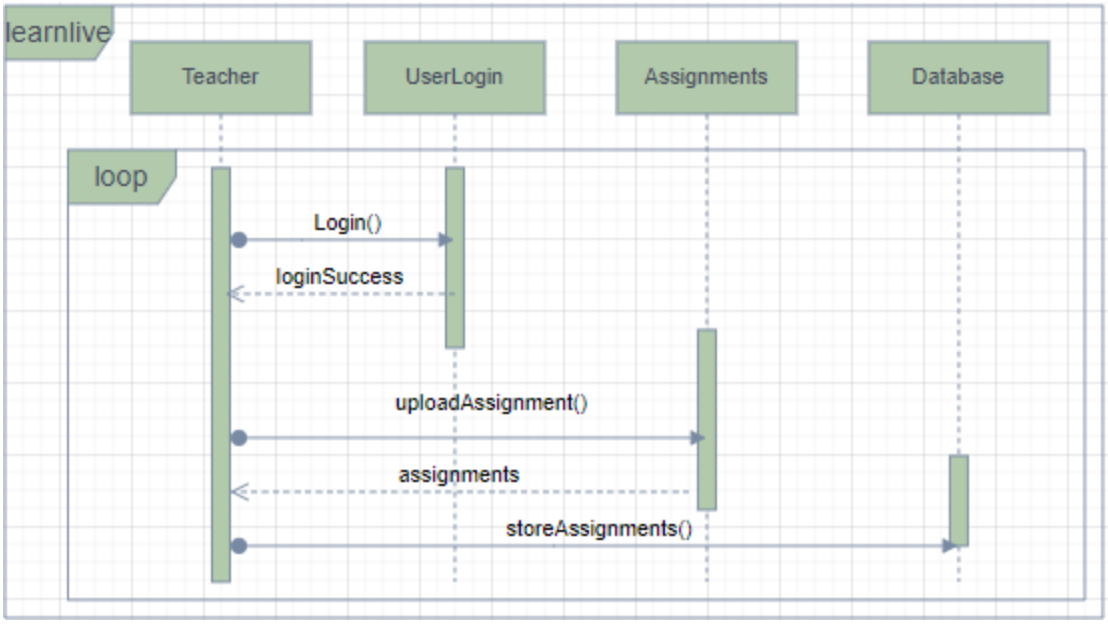
Operation:	Generate Certificate
Cross References:	Assign Student
Pre-Conditions: • The student has been assigned to a camp by the admin • Admin can monitor the camp details	
Post Conditions: • Student completes the duration of camp • Admin generates the certificate upon completion • Student can see his certificate at his side	

5.4.6 Contract Use Case : UC-17

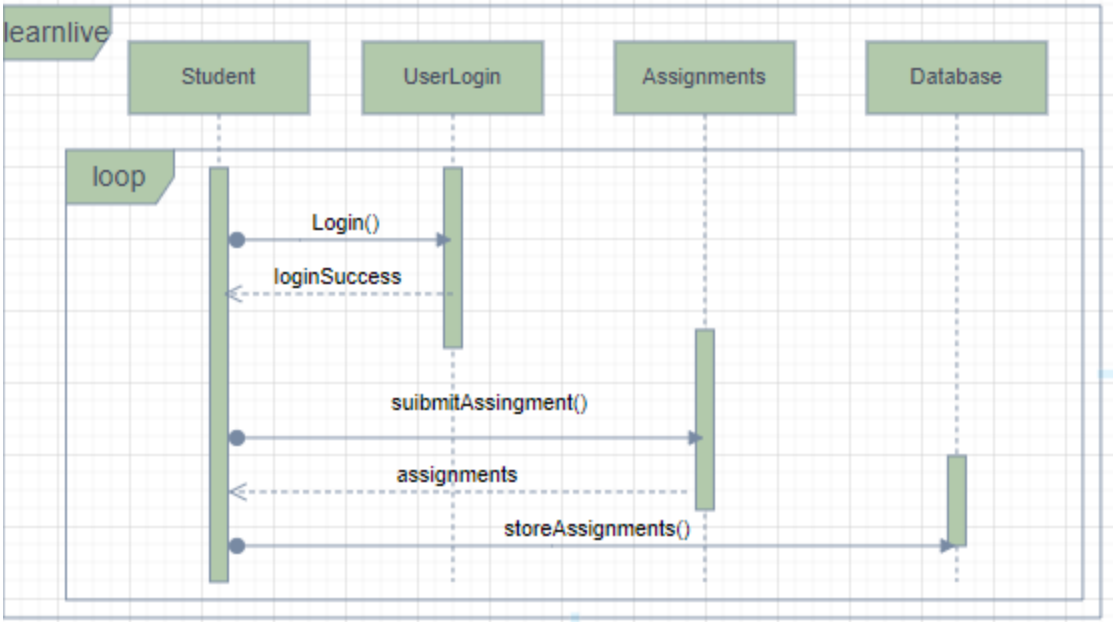
Operation:	Collaborate Code
Cross References:	Write Code
Pre-Conditions: • The code has already been written by student	
Post Conditions: • Student creates the room for teacher to join in • Student and teacher are now in the same room • Student and teacher can collaborate for coding	

5.5 Sequence Diagrams

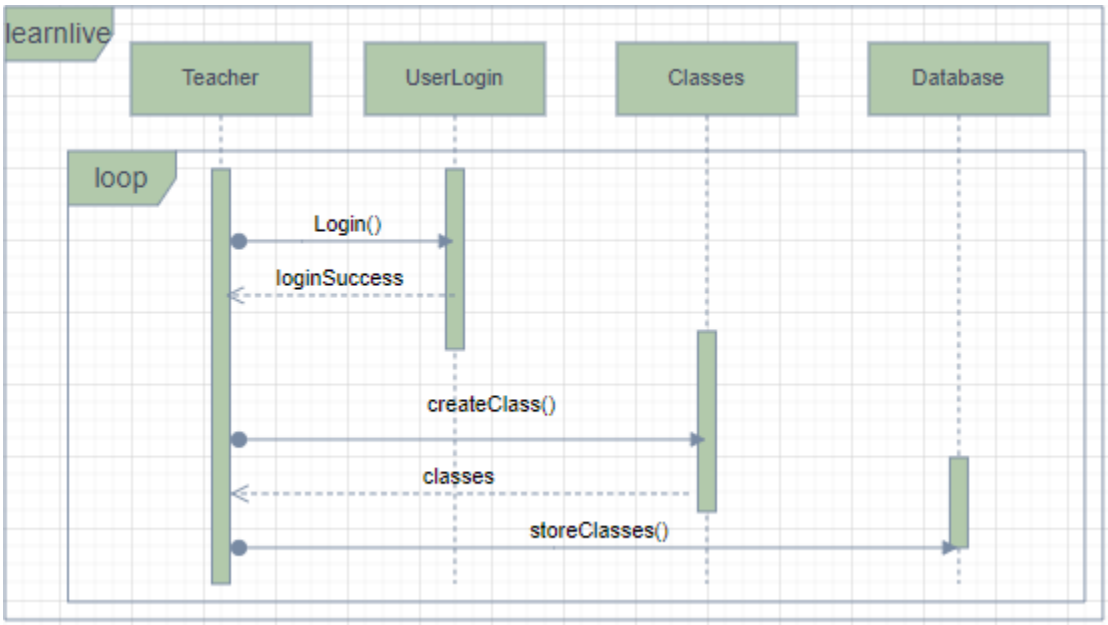
5.5.1 Use Case 3 : Upload Assignments



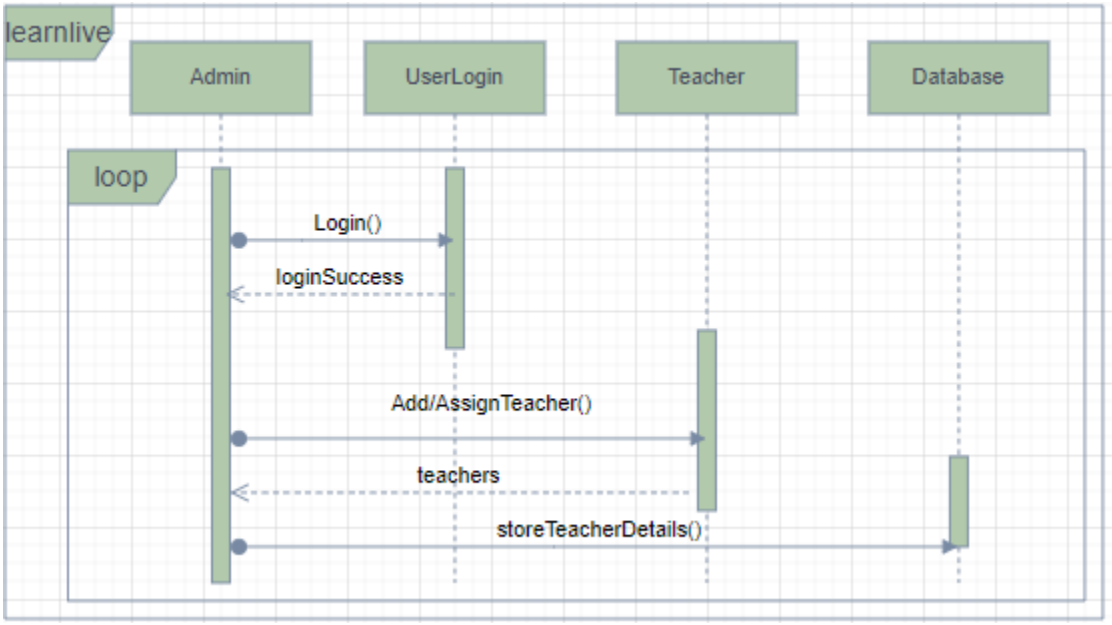
5.5.2 Use Case 5: Submit Assignments



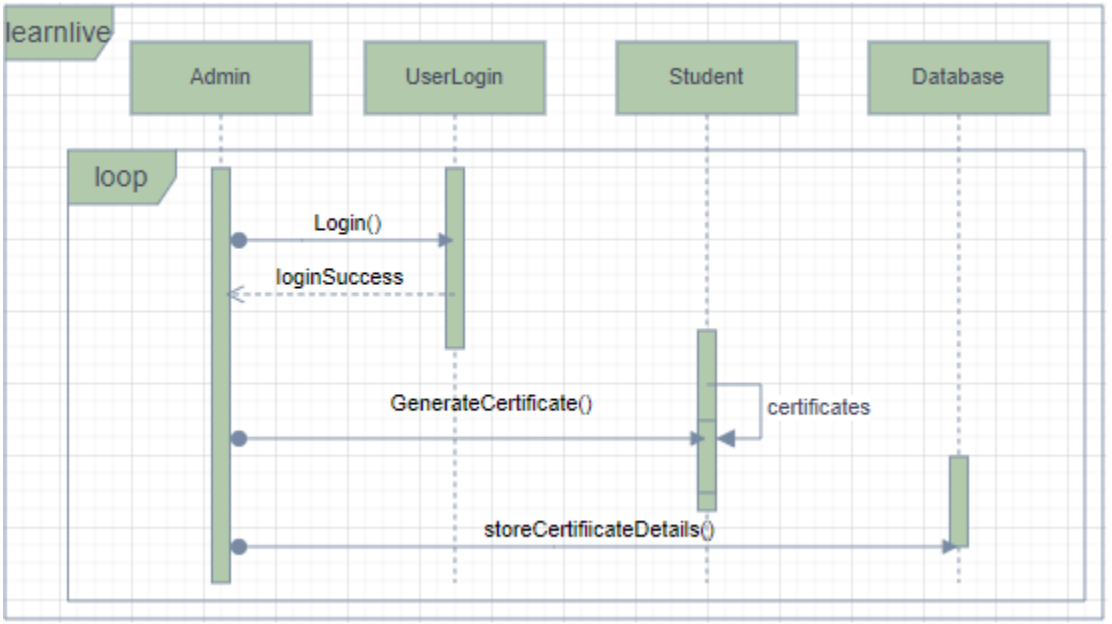
5.5.3 Use Case 10: Create Class



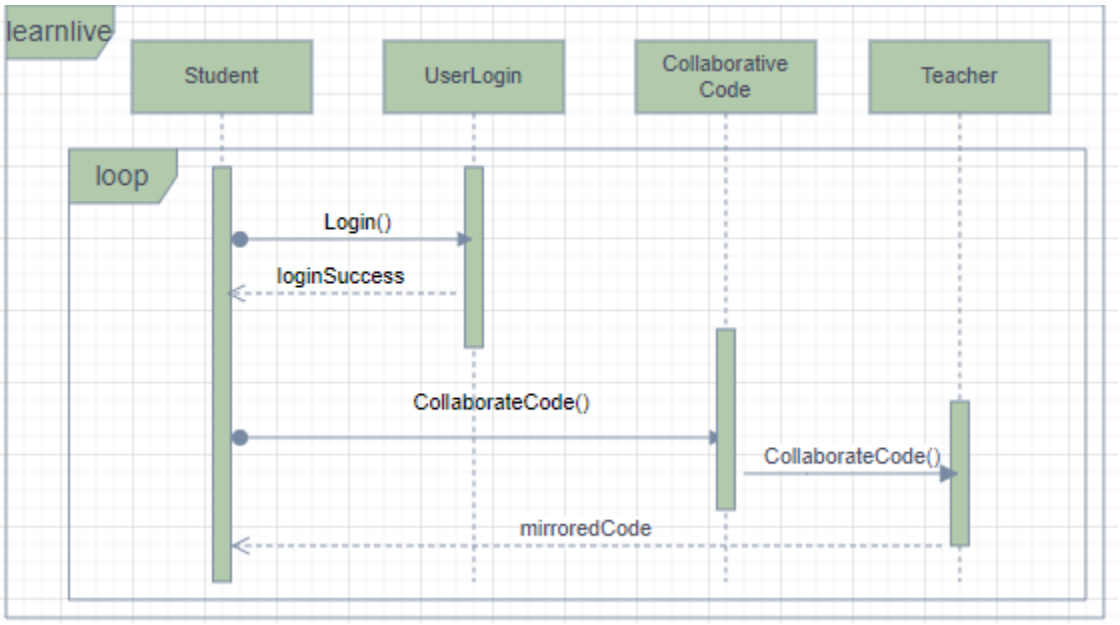
5.5.4 Use Case 11: Add/Assign Teachers



5.5.5 Use Case 14: Generate Certificate



5.5.6 Use Case 17: Collaborate Code



5.6 System Sequence Diagram

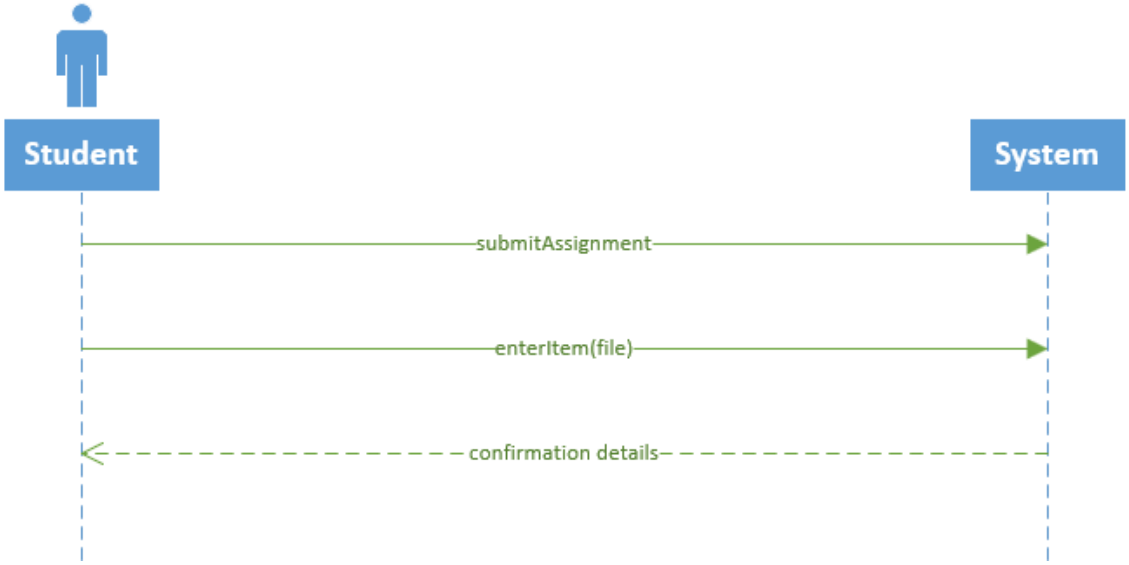
5.6.1 Upload Assignment



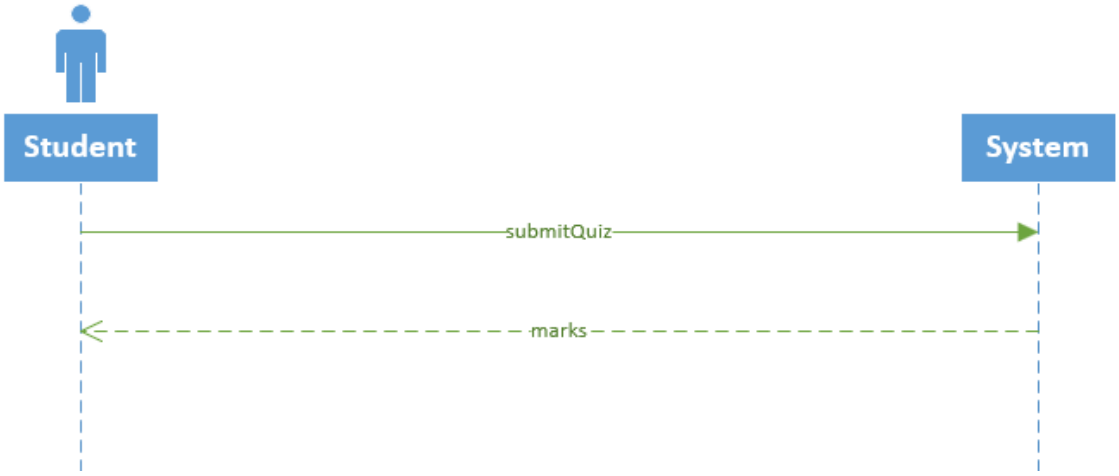
5.6.2 Upload Quiz



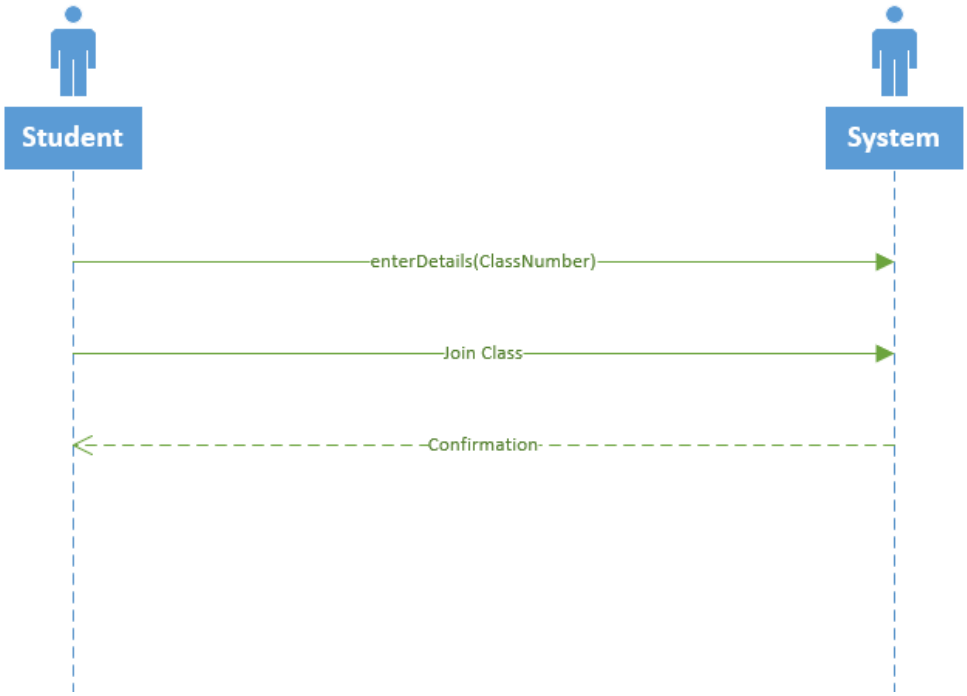
5.6.3 Submit Assignment



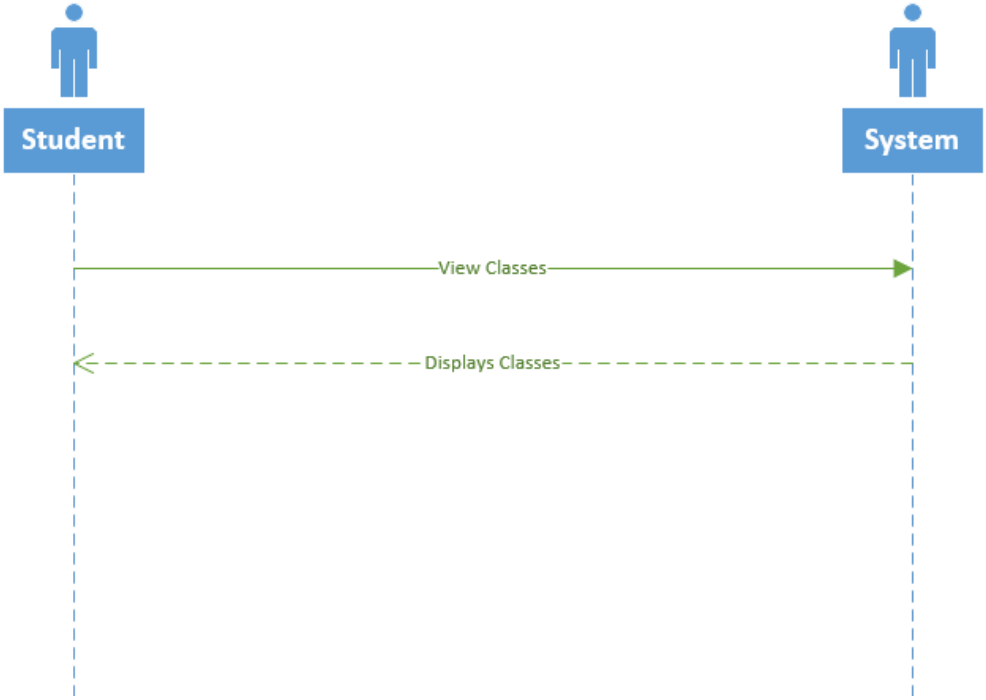
5.6.4 Submit Quiz



5.6.5 Join Class



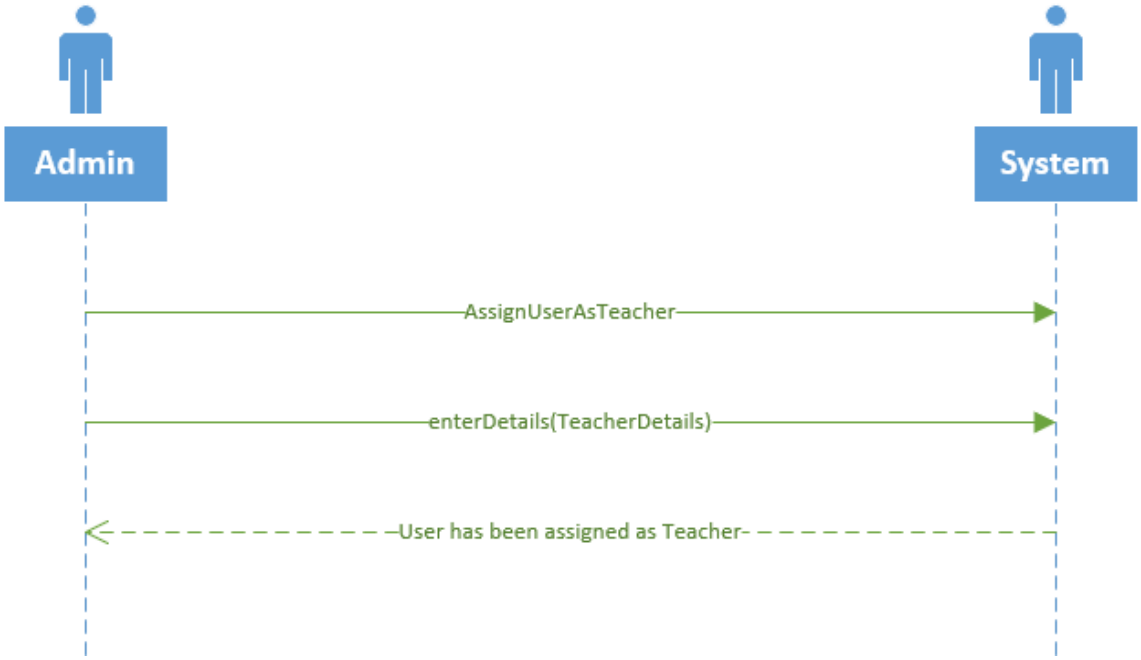
5.6.6 View Classes



5.6.7 Create a Class



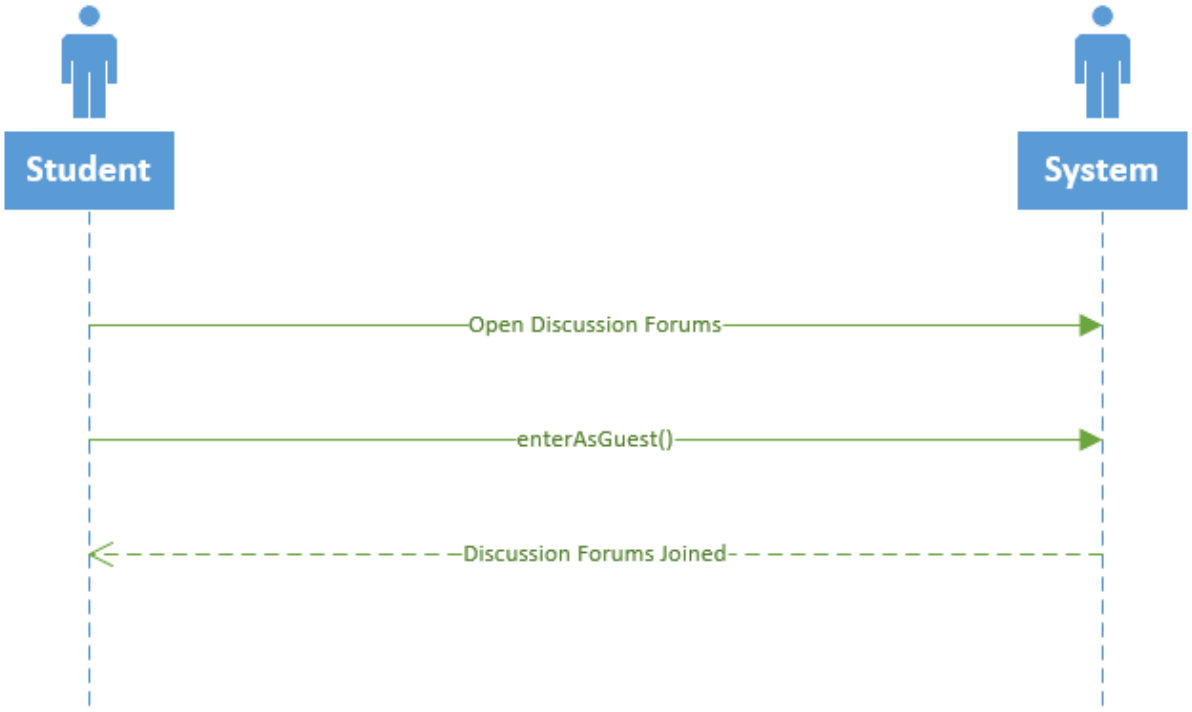
5.6.8 Assign Teacher



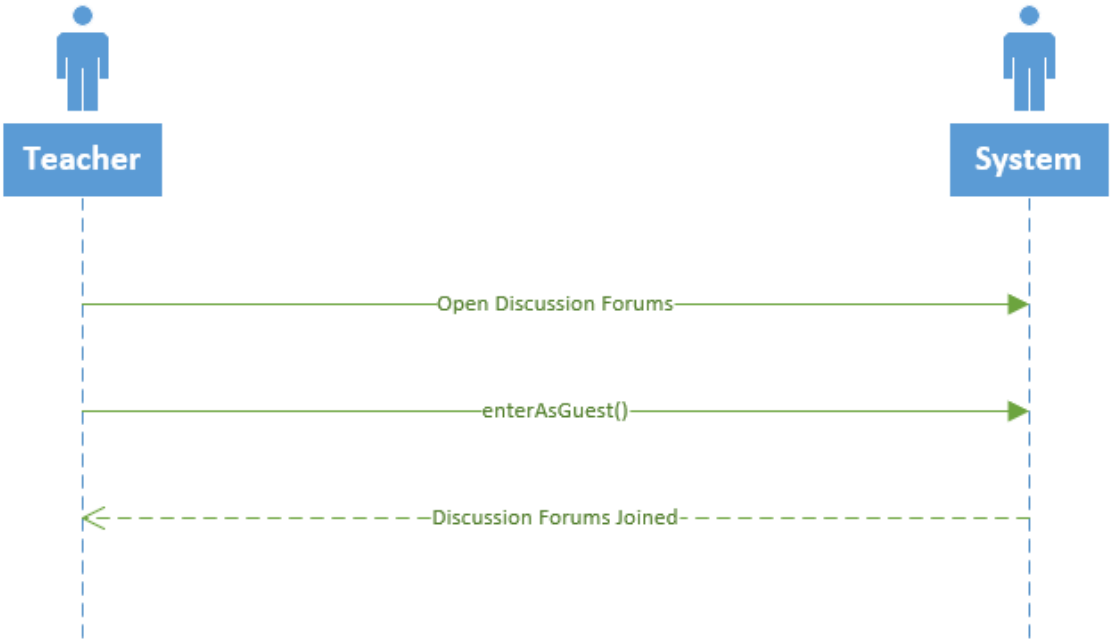
5.6.9 Assign Student



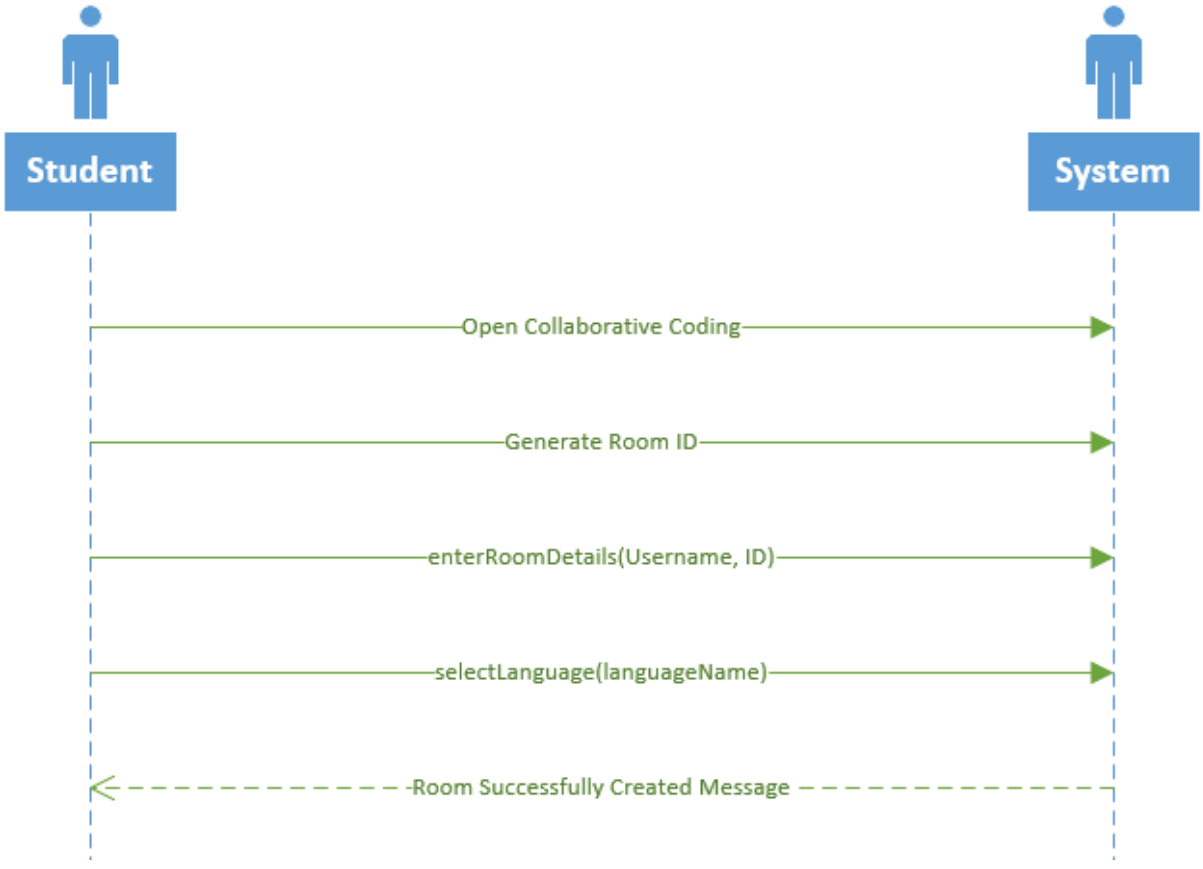
5.6.9 Student Joins Discussion Forums



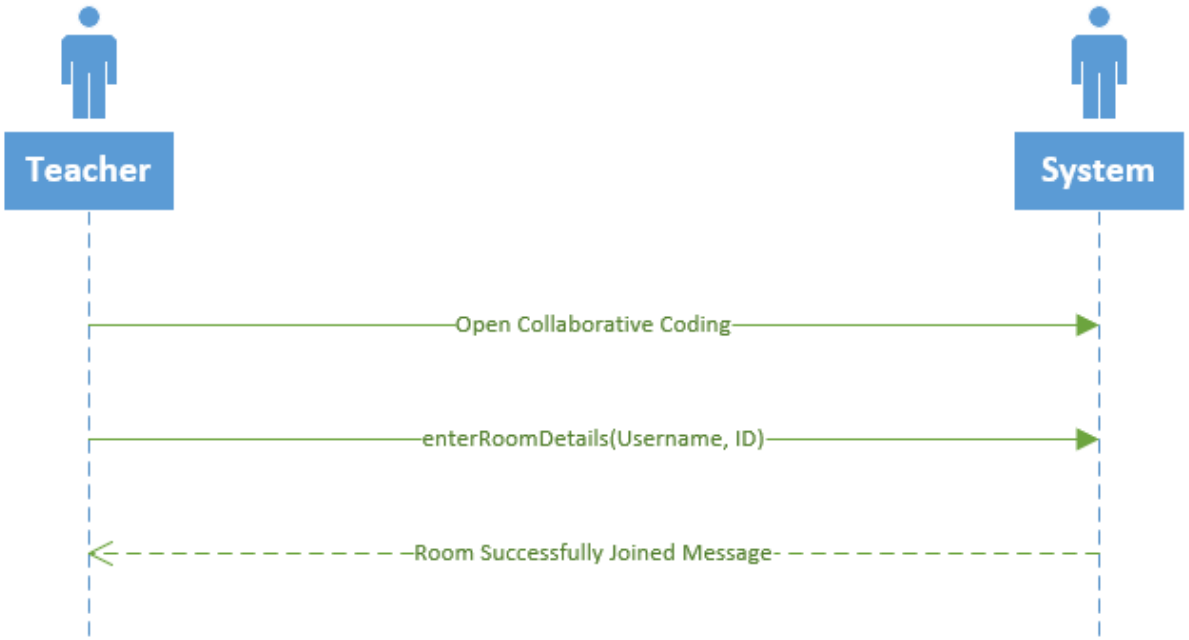
5.6.10 Teacher Joins Discussion Forums



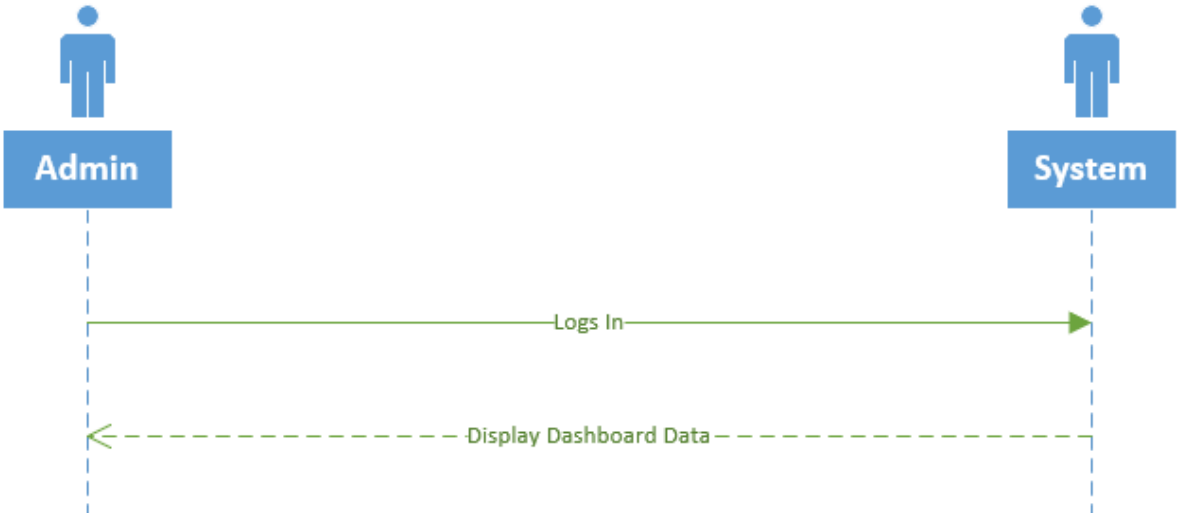
5.6.11 Student Creates Collaborative Coding Room



5.6.12 Teacher Joins Collaborative Coding Room

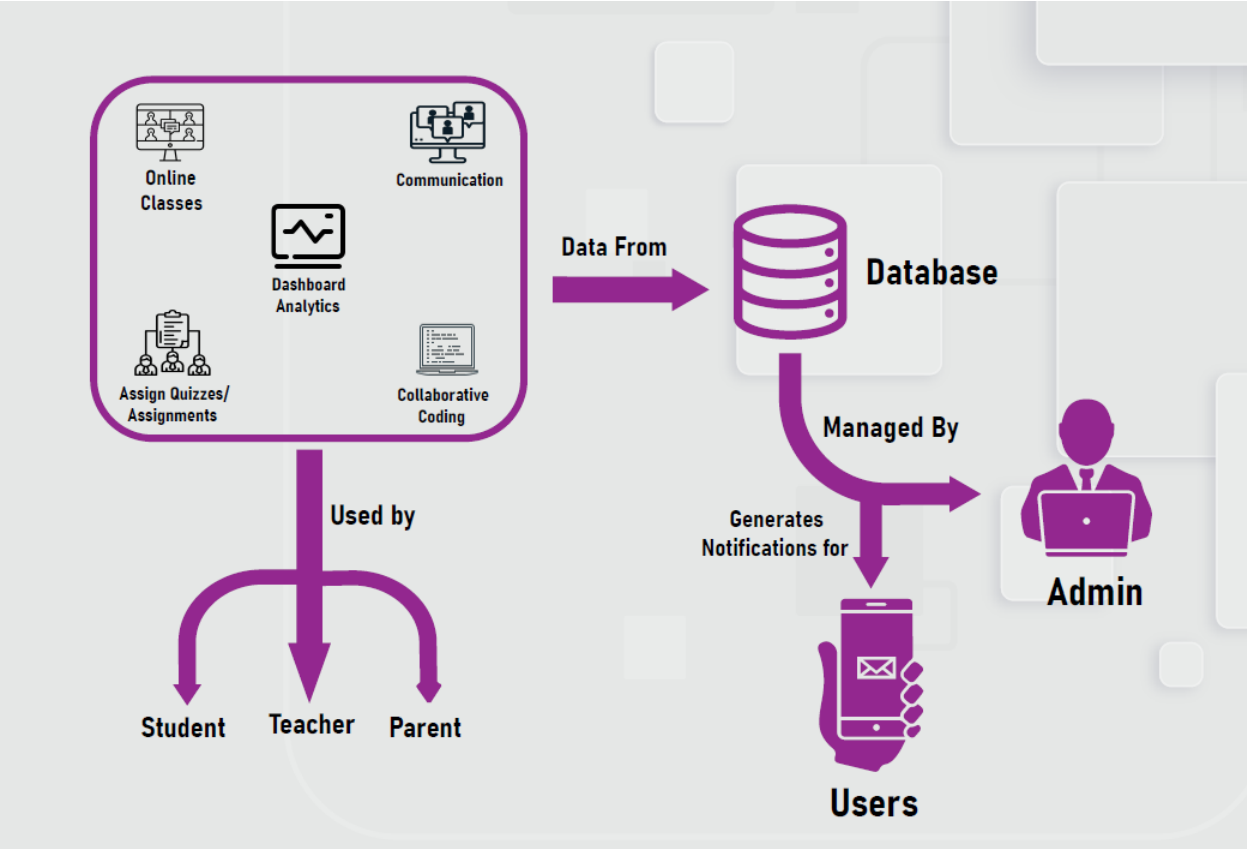


5.6.13 View Dashboard Analytics

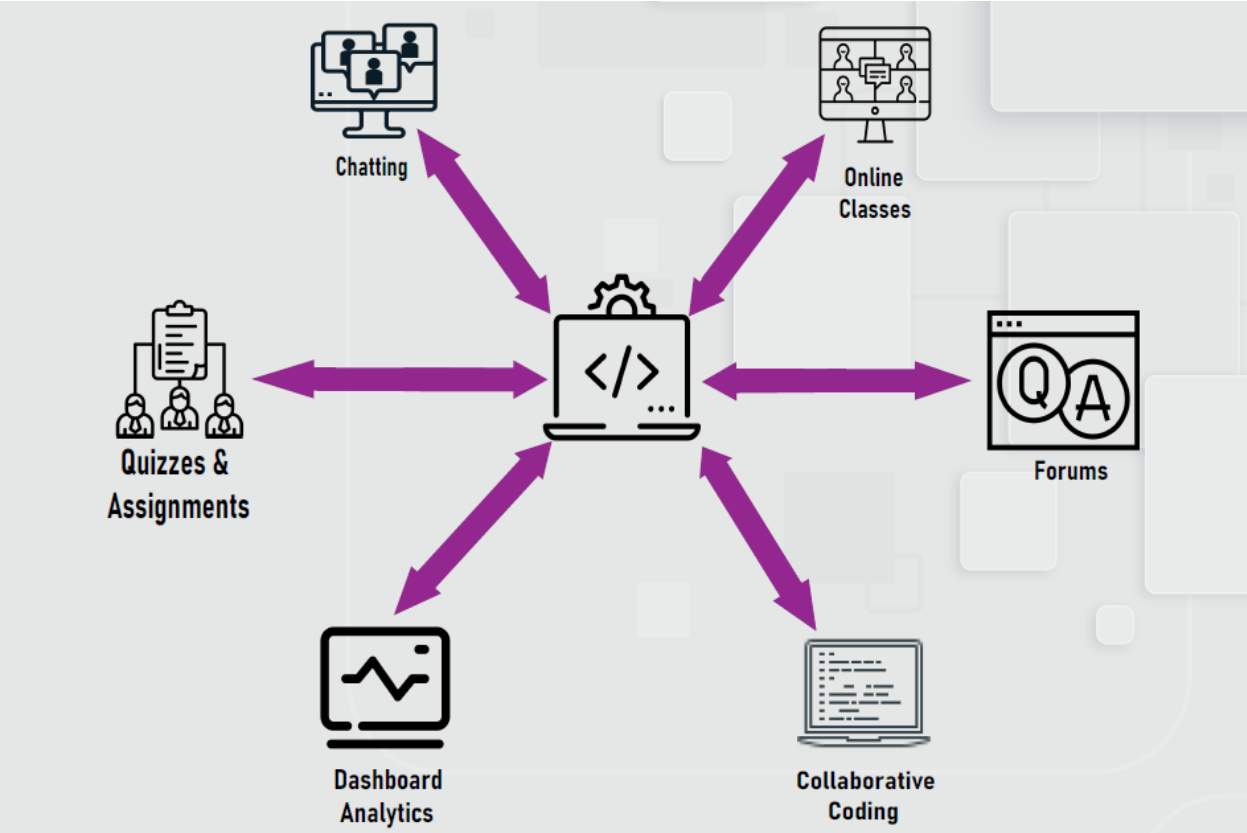


Chapter 6: Design

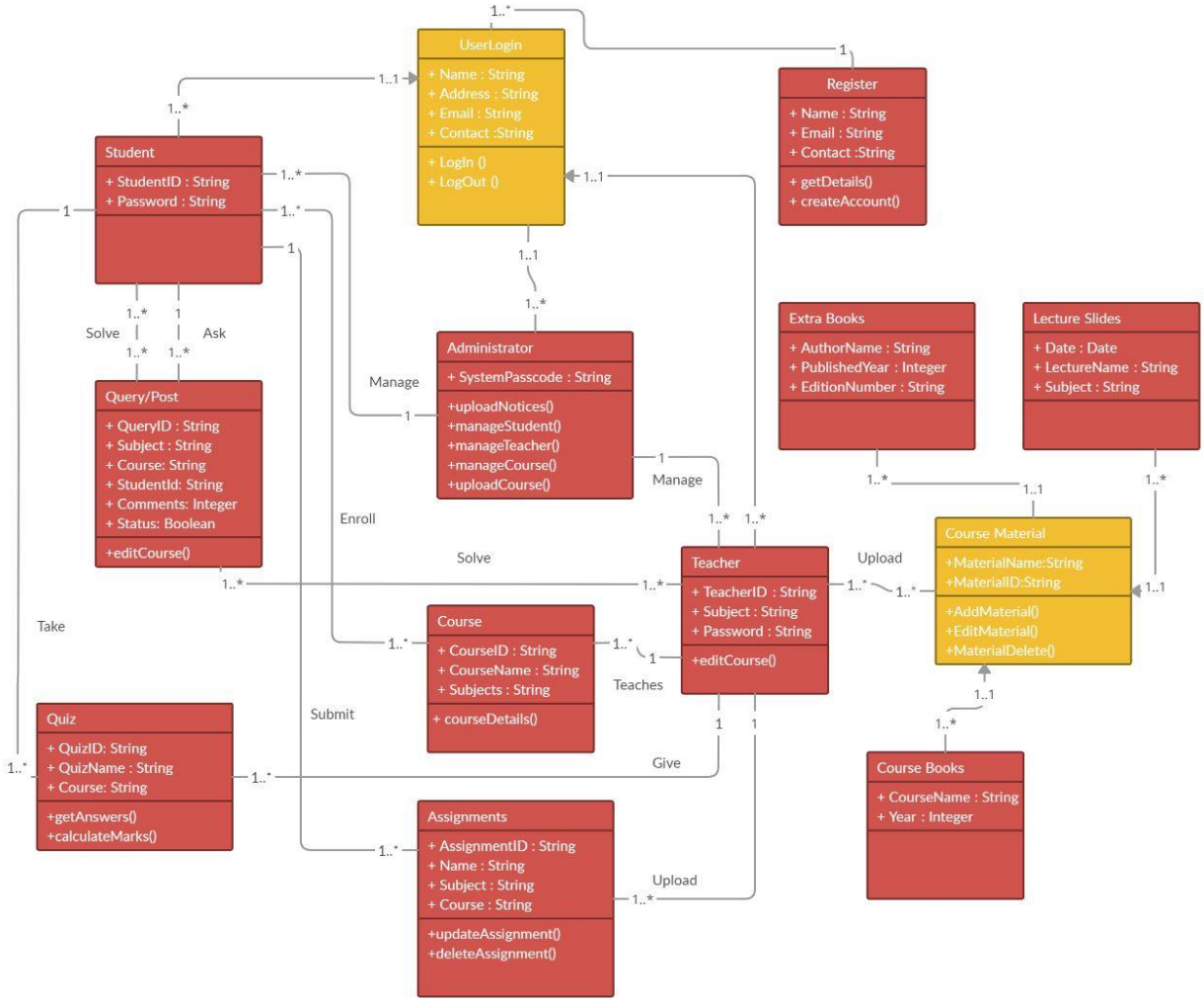
6.1 Architecture Diagram



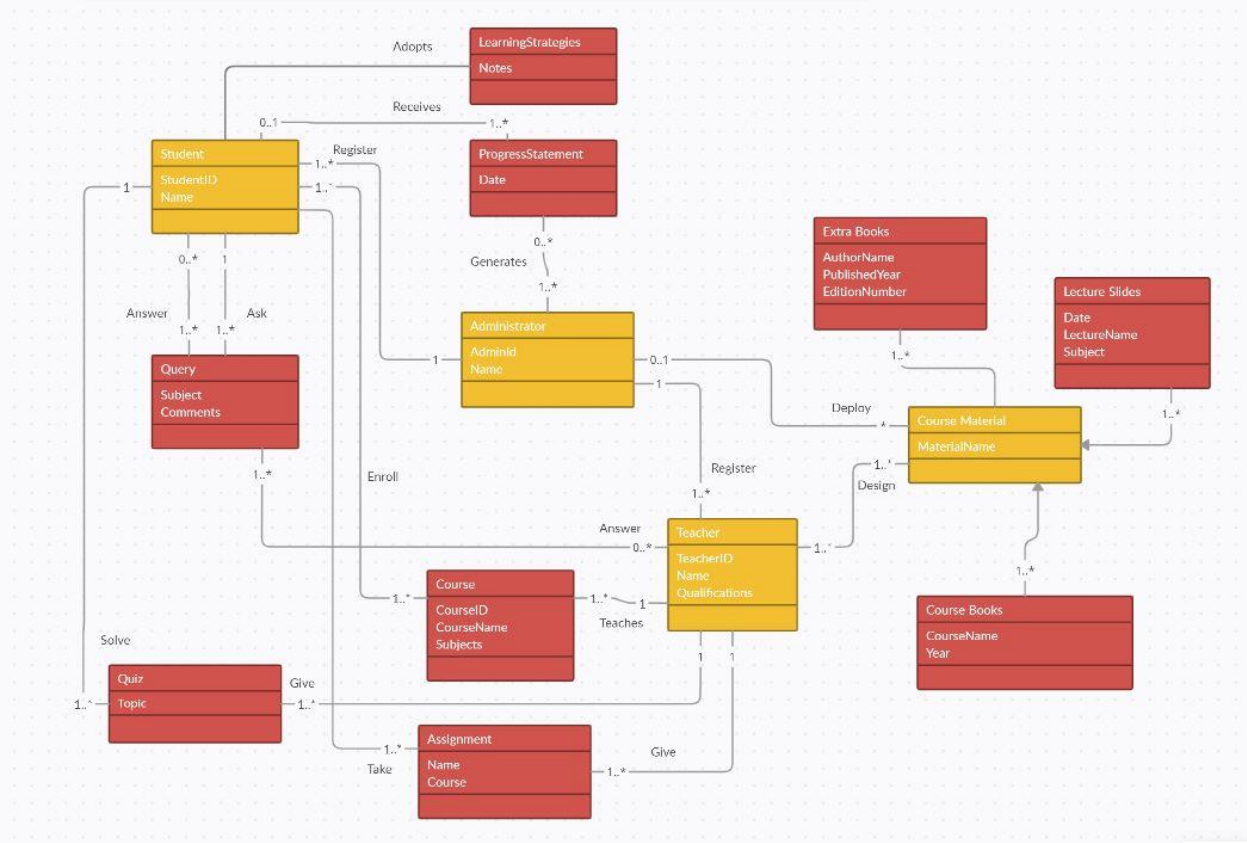
6.2 Workflow Diagram



6.3 Class Diagram



6.4 Domain Model Diagram



Chapter 7: Iteration Plan

7.1 FYP-1

- Detailed UX design with hi-fi prototype of the entire concept
- Profiles for Teachers, Parents and Students

Admin Panel:

- Schedule Classes (no conflicts) and also this will become the timetable to be displayed weekly or monthly (calendar)
- Assigning learners to the camp
- Assigning Teachers to the Camps
- Assigning Managers or Support Team
- Control Panel for setting the permissions for all
- Notifications about each class being started, teacher and student joined
- Class starting, ending time and attendance log and active hours of each participant
- Notifications of task or weekly challenge or Project being released and if not then a reminder for the teacher to do it
- Notifications of authenticated all slides for exhibition and if not then a reminder for the teacher to do it
- Notifications about certificate being ready and if not then a reminder for the teacher to do it

Working Space for Learners:

- They can join a scheduled class, early reminders and notifications for joining late are included
- Attending the class and participating as set by the teacher
- Developing class tasks, weekly challenges or projects in the platform which are shareable to friends and family for reviews or ratings
- Performing quizzes
- Access to digital certificate
- Slide deck for exhibition
- Access to join the live exhibition
- Coding class with provision of 7 students in a class, one teacher, one manager and one executive. Summing it all make 10 users in a class with varying controls and rights. Access to students for peer review or peer task
- Tracking of students' attendance and active time
- Interactive quiz in the class by logging results and progress

7.2 FYP-2

- **Chat/Channels/Forums:**
    - Chat between parents and executives

    - Chat between teachers and Camp Manager and executives
    - Chat between Camp Manager and executives
- **Working Space for Parents:**
    - Class schedule as calendar invites of each day
    - Class reminder to the parents
    - Notification to the parent if learner has joined or not
    - Provision to send message or call to parent by Camp Manger or executives during class if needed
    - Provision of notes for parents by teachers to be followed up by Camp Manager or Executives further
    - View the class task, weekly challenge and Project being set for the learners, they can comment or share feedback for everything
    - View Leaders board and access to the detailed evaluation and feedback to identify the gaps
    - View Certificate
    - Share their Parent review as a video or text message and share rating for the camp, teacher and team
    - Join Live Exhibition
- **Workbench for Teachers:**
    - Joining Scheduled class, early reminders and notifications for joining late included
    - Designing Lesson Plans using Templates and utilizing previous resources
    - Designing and providing class tasks, Weekly Challenges and Projects with Checkpoints and score scheme
    - Preparing Leaders board through evaluation to be accessible to parents and students
    - Preparing interactive quizzes
    - Teachers can share the student's work to social media handler
    - Code Reviews
    - Authenticating Certificates to the learners of the camp
    - Authenticating and reviewing exhibition slide deck
    - Joining the live exhibition
    - Learning analytics from the class (dashboard)

Chapter 8: Experimental Results and Analysis

This chapter includes information related to the analysis done while choosing the technologies i.e. front and back end technologies to be used to develop the system Shopinect. Supervised by an external supervisor, we did not get the opportunity of experimenting and testing different algorithms and technologies before using them in our project. Since it is an industrial project, many decisions during the development process depended on external supervision rather than experimental analysis done by us.

However, we did get a chance to learn and study different front-end and back-end frameworks before embarking on boarding the project.

Following are results of our research:

8.1 Front-End

8.1.1 Choosing ReactJS

- It uses Javascript and JSX programming language, the same as the backend so we don't have to learn a different programming language for both backend and frontend.
- ReactJS is extremely intuitive to work with and provides interactivity to the layout of any UI. Plus, it enables fast and quality assured application development that in turn saves time for both - clients and developers.
- It helps us build modular, component based code upon and gives enough tools to take away the very manual work it would take to do it in regular Javascript.
- Compared to other frontend frameworks, the React code is easier to maintain and is flexible due to its modular structure. This flexibility, in turn, saves a huge amount of time and cost for businesses.
- React JS was designed to provide high performance in mind. The core of the framework offers a virtual DOM program and server-side rendering, which makes complex apps run extremely fast.
- One of the main benefits of using React JS is its potential to reuse components. It saves time for developers as they don't have to write various codes for the same features.

8.1.2 Reasons not to choose AngularJS

- The variety of different structures (Injectable, Components, Pipes, Modules, etc.) makes it difficult to learn as compared to React and Vue.js, both of which consist of just one component.
- Angular is relatively slow. Even while it may be addressed by employing the so-called "Change Detection Strategy," which aids in the human control of component rendering, it does not significantly improve performance.

8.2 Back-End

8.2.1 Choosing NodeJS

- NodeJS eases handling multiple requests made by the client and enables code sharing and re-usage of library codes. The apps get immediate responses and work within the limited time frame. Its single-threaded functionality makes it best suited for real-time communication.
- You can easily build APIs that are responsive, efficient, and lightweight.
- Making Single Page Applications is possible.
- It is supported by a large community of developers that are constantly working on improving it. It performs high-level data streaming.
- Node.js handles simultaneous tasks, since it is non-blocking and supports async functions.
- Node.js backend development is concise, as it provides many features that can be impossible to have with any event-based environments.

8.2.2 NodeJS Disadvantages

- Reduces performance when handling Heavy Computing Tasks
- Node.js invites a lot of code changes due to Unstable API
- Node.js Asynchronous Programming Model makes it difficult to maintain code

Chapter 9: Implementation

Ask the author of this repository for .env files which cannot be downloaded directly from the repository due to security reasons. Afterwards,

Open the folder in 2 terminals side by side and write these commands

Terminal 1:

nd backend
npm install
npm start

Terminal 2:

nd frontend
npm install
npm start

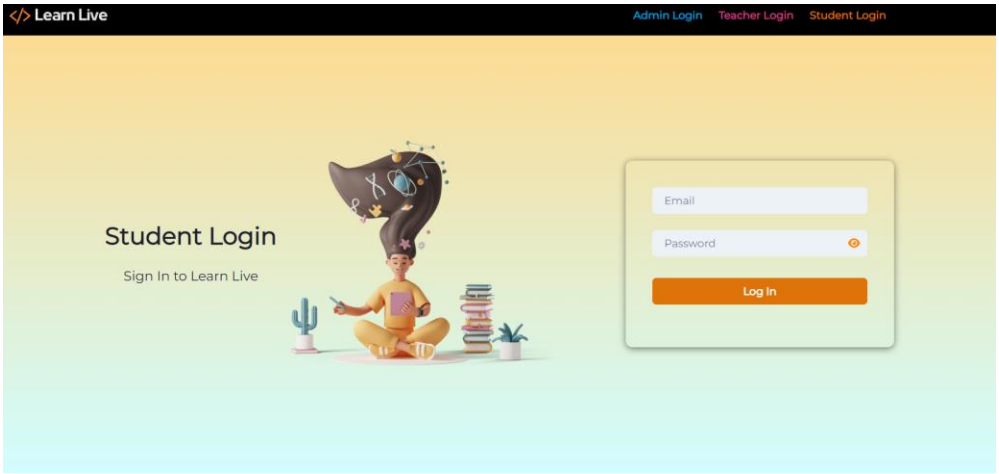
After pressing enter, you will be directed to main page of our application.

9.1 Main Page

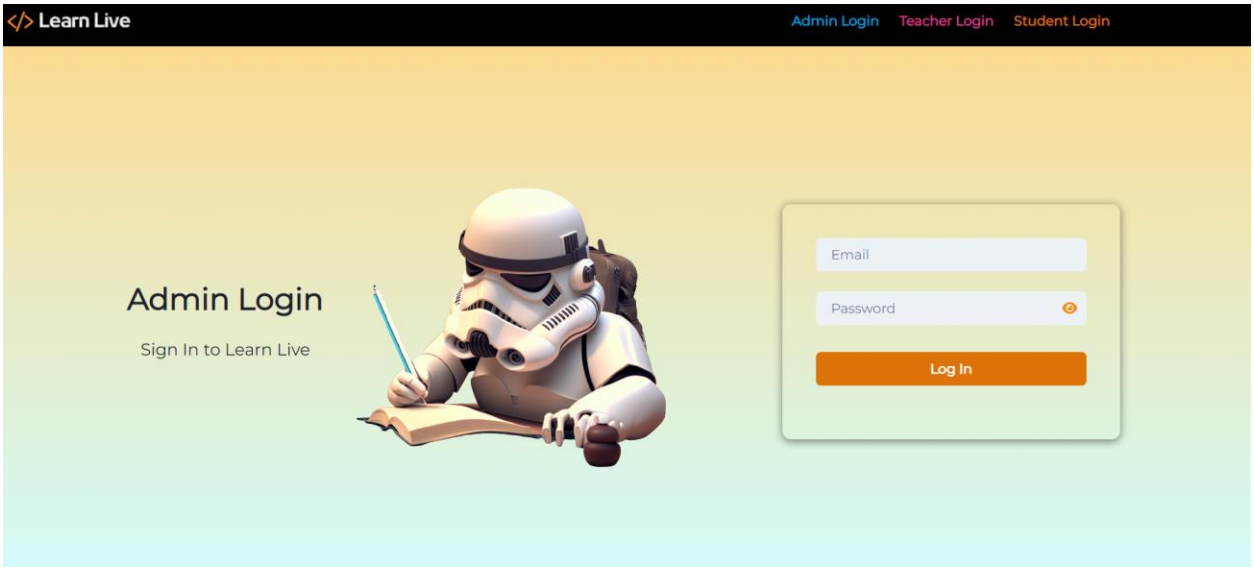
We have multiple profiles

- Admin
- Student
- Teacher

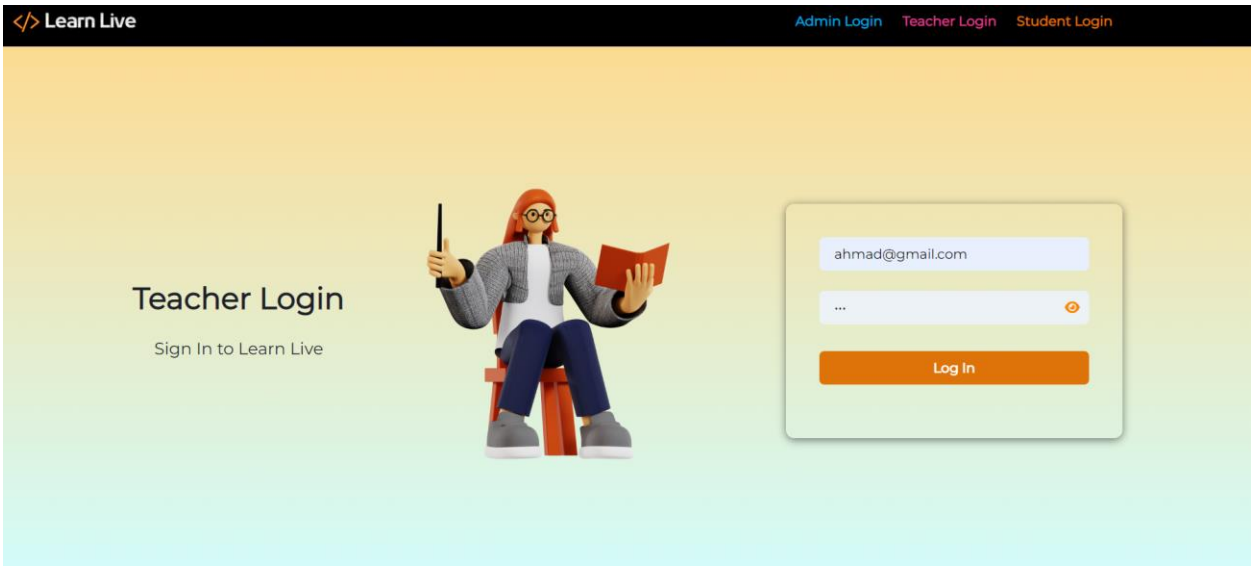
9.1.1 Student Login



9.1.2 Admin Login



9.1.3 Teacher Login



9.2 Admin Panel

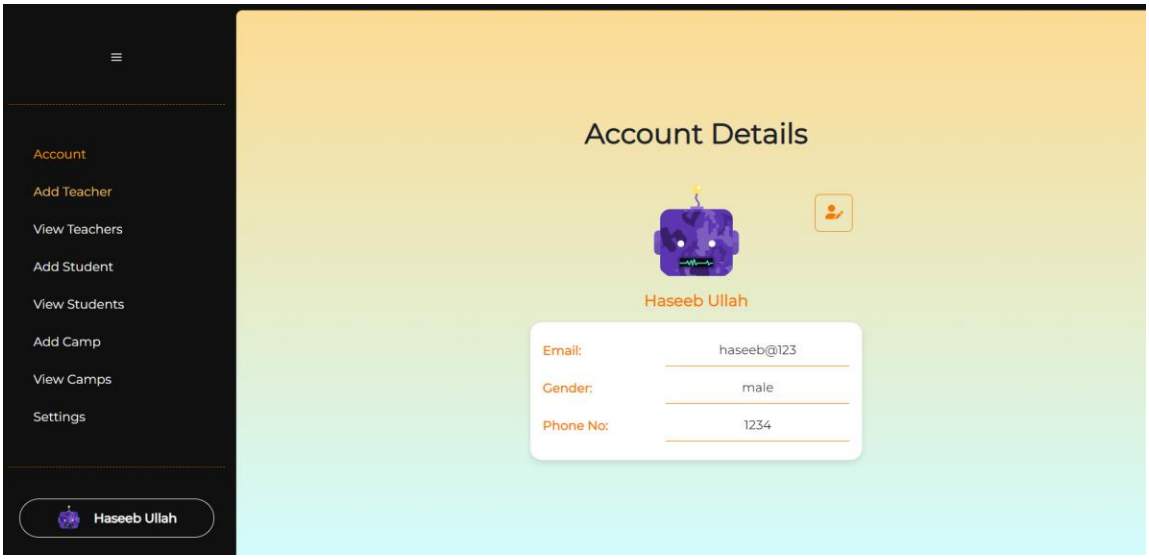
Only admin can login into this.

Main responsibilities of Admin are:

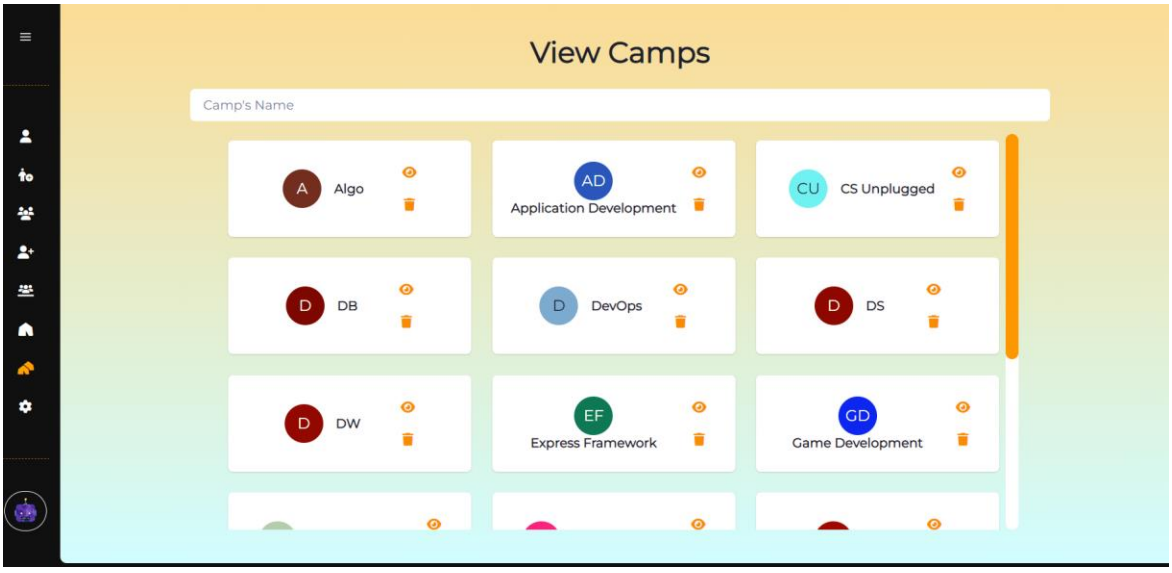
- Add Students and assign students in camps
- Add Teachers and assign teachers in camps

- Add Camps
- View Camps
- Generate certificate for students at camp completion

9.2.1 View Account Details

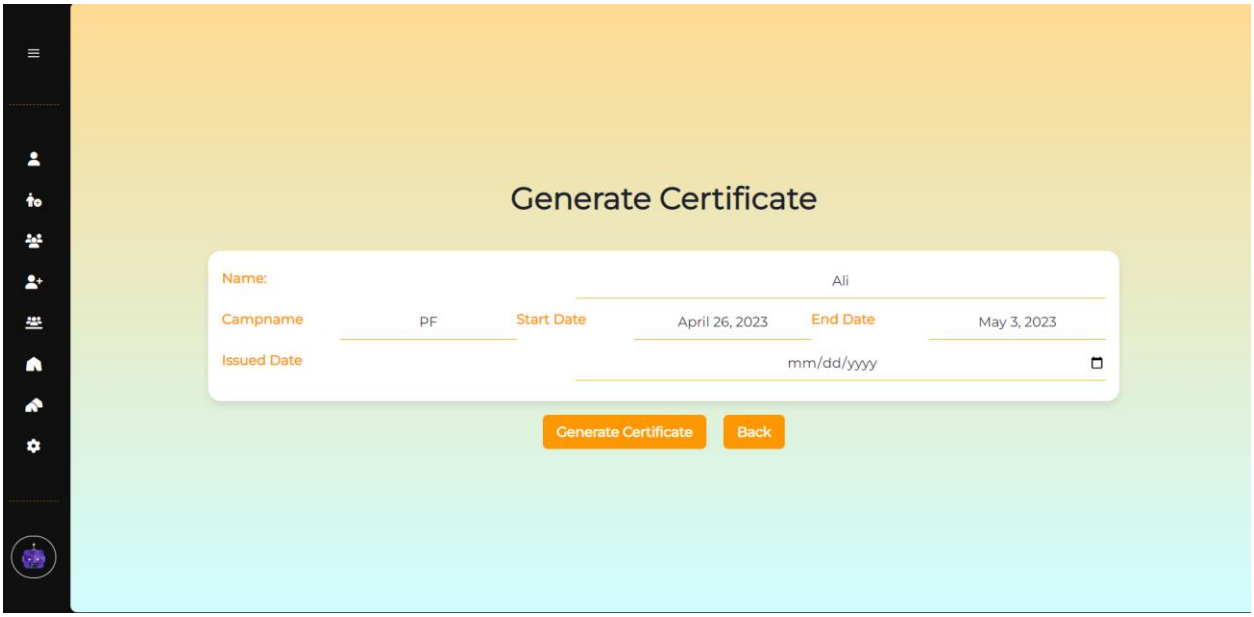


9.2.2 View Camps

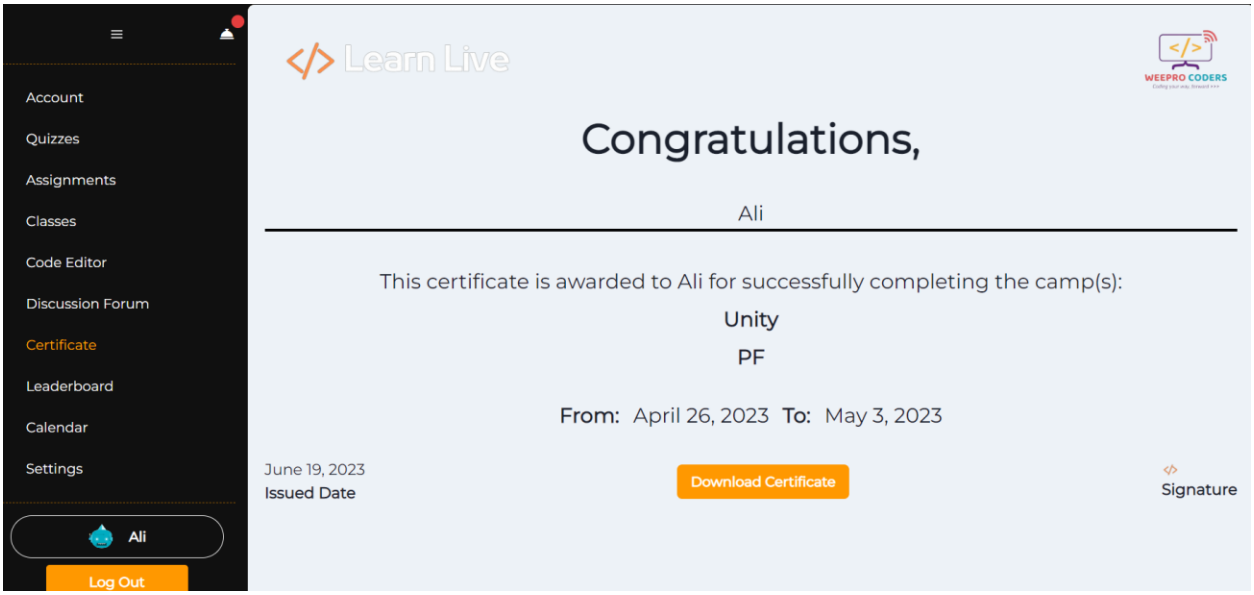


9.3 Generate Certificate

Admin can generate certificates for the student at the end of camp.



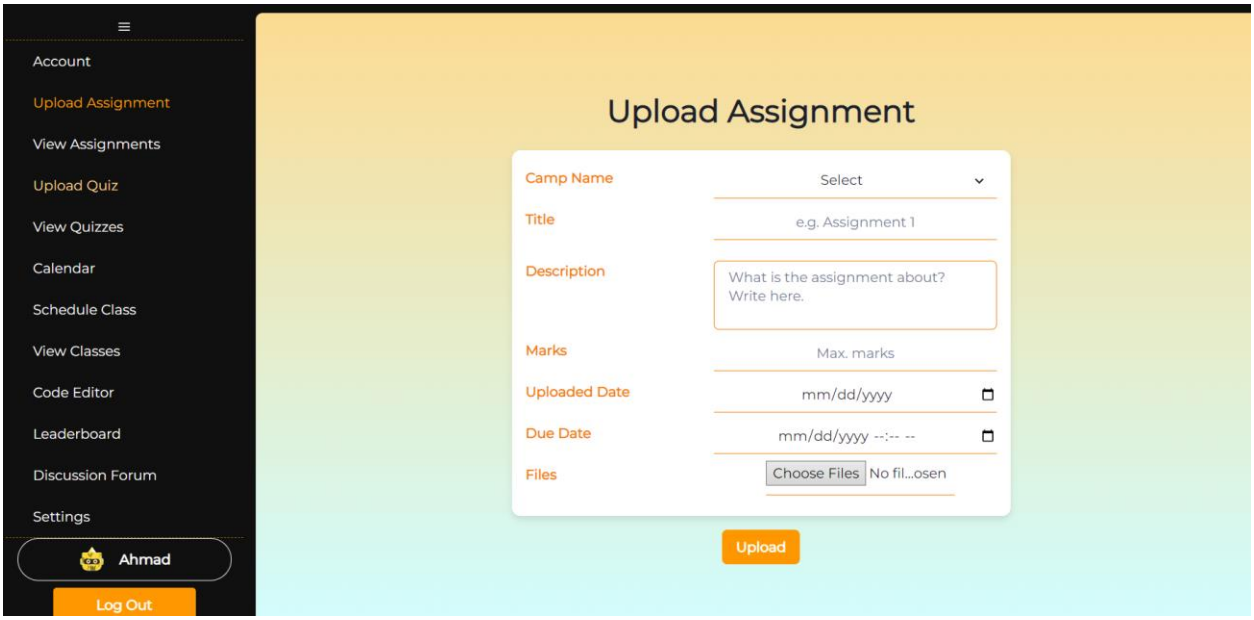
Which can be seen on Student side



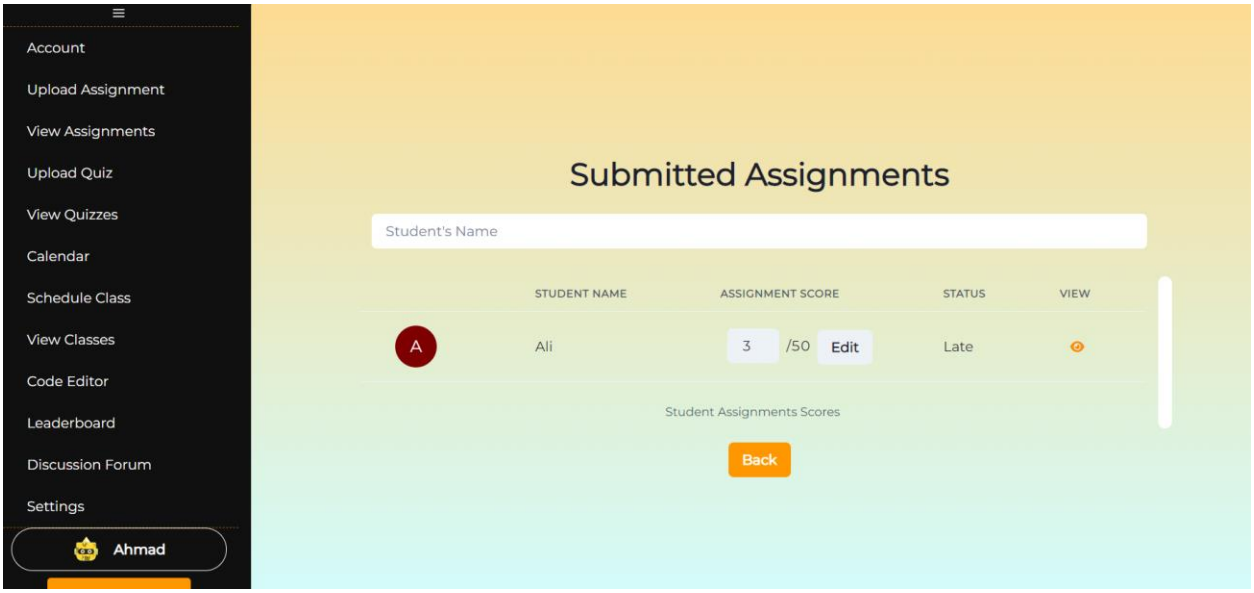
9.4 Upload/Viewing Assignments and Quizzes

9.4.1 Upload/Viewing Assignments

Teacher is able to upload assignments.

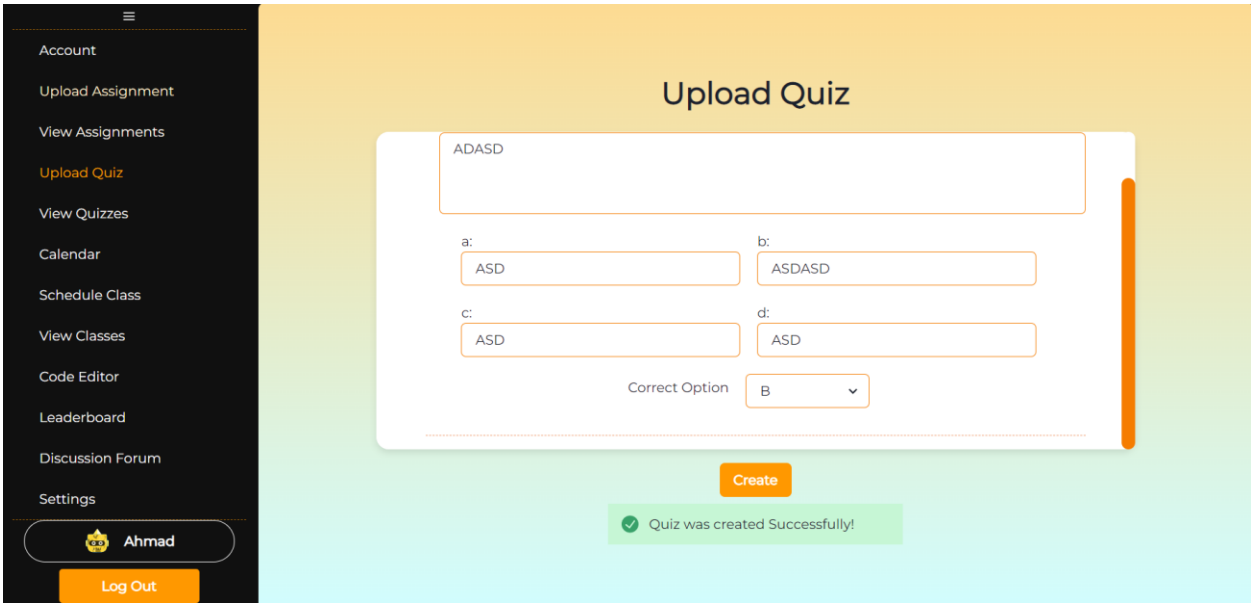


Teacher is able to see submissions also.

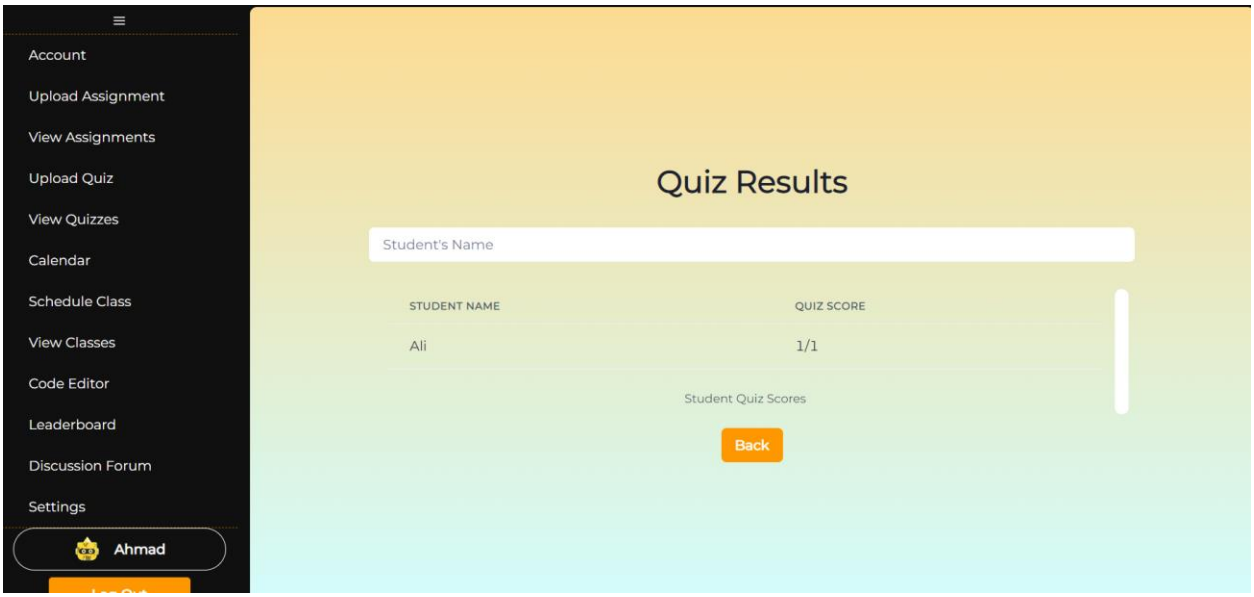


9.4.2 Upload/View Quizzes

Teacher is able to upload quizzes.



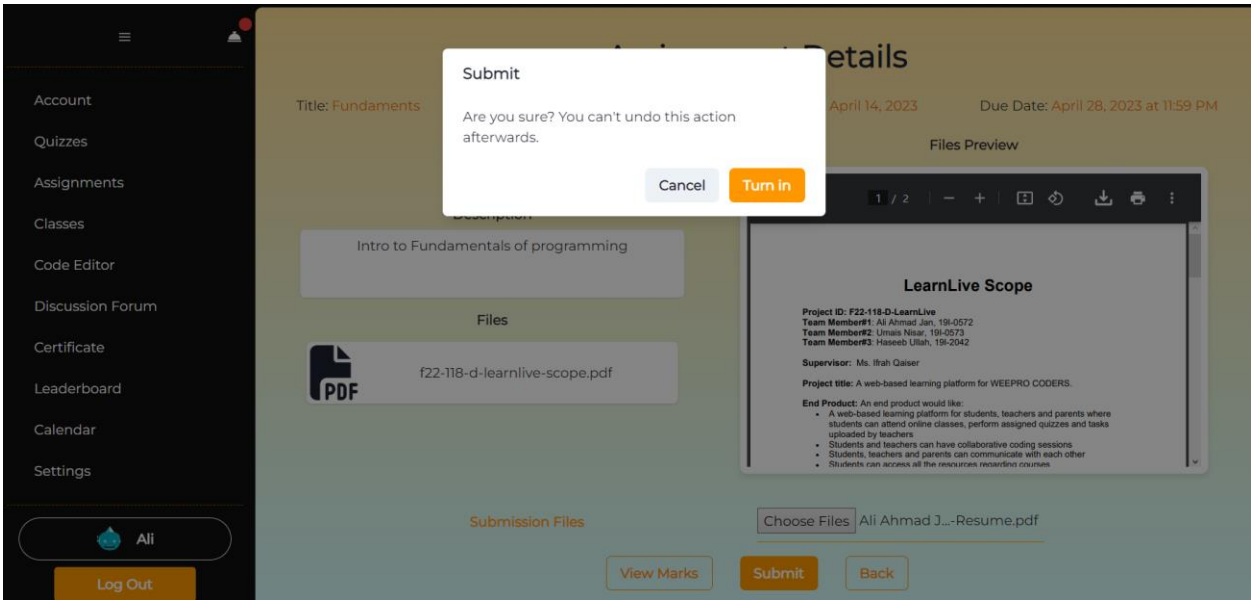
Teacher is able to see result of specific quiz.



9.5 Submitting Assignments and Quizzes

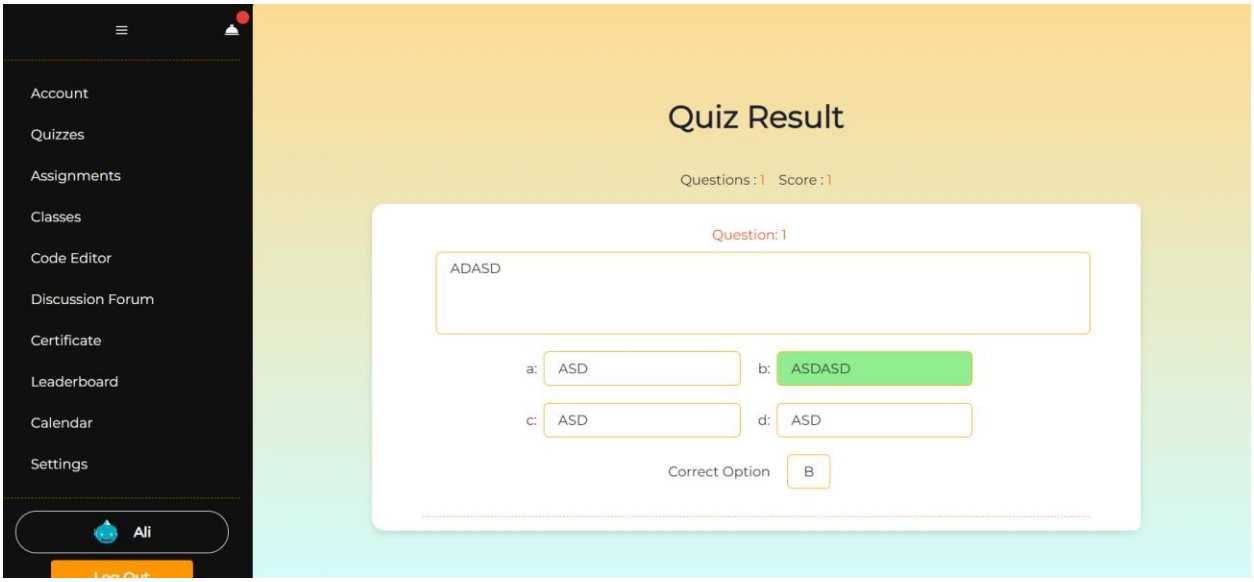
9.5.1 Submit Assignment

Students are able to turn in assignments after attaching them.



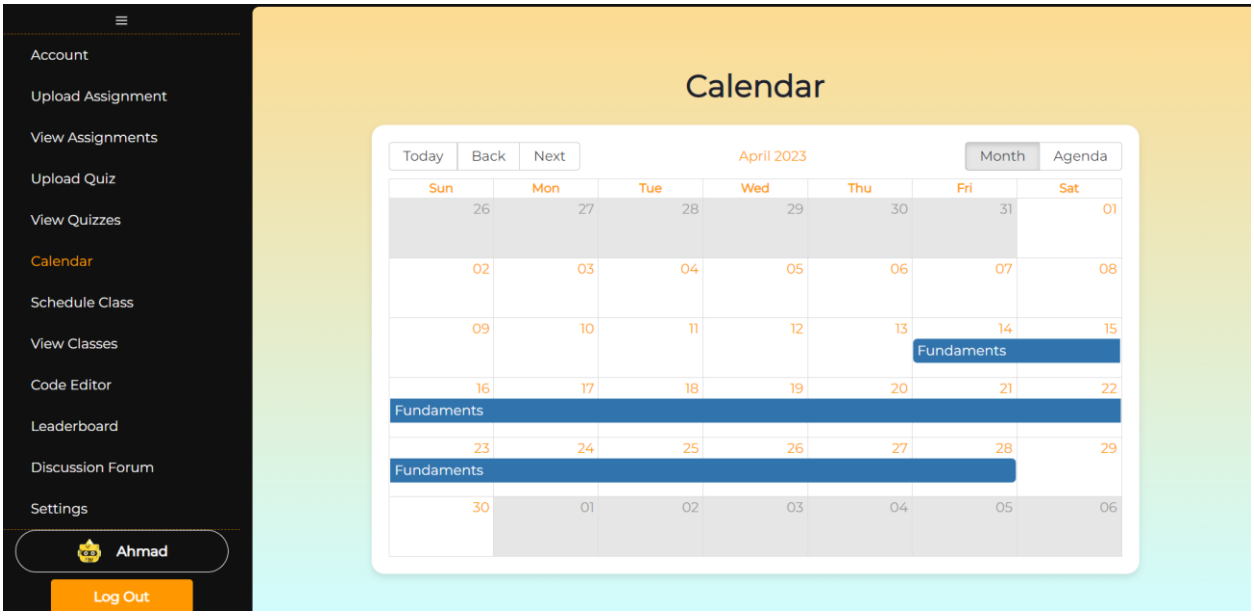
9.5.2 Submit Quiz

Students are able to submit the quiz and see the score as well.



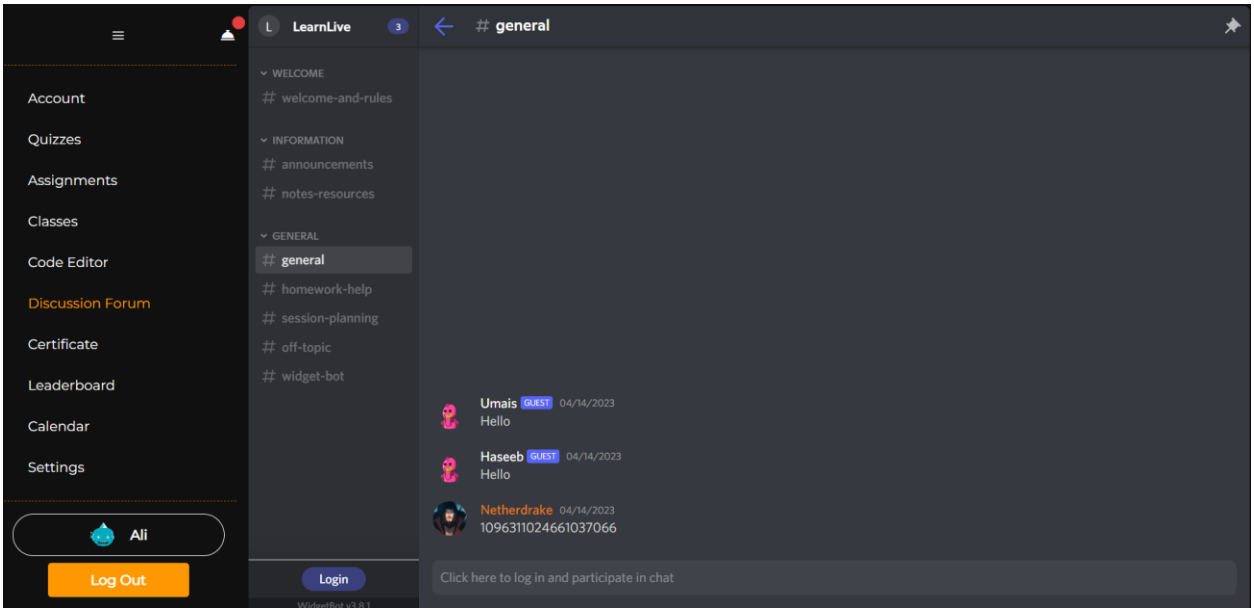
9.6 Calendar

Both teacher and student can see assignment submission dates on calendar



9.7 Discussion Forum

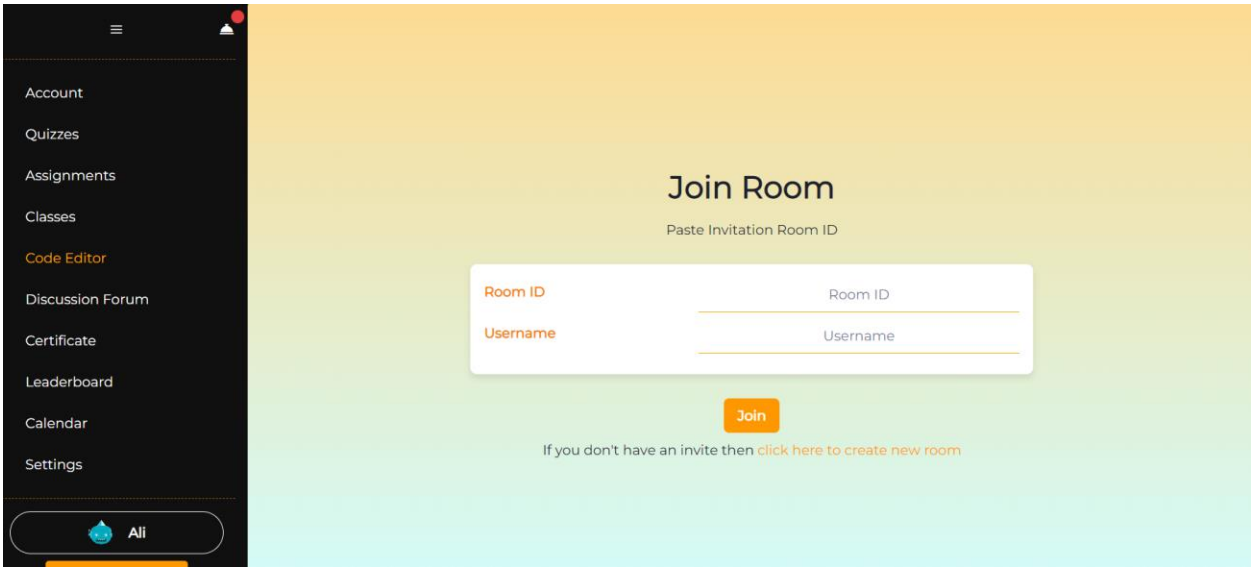
Both teacher and student are able to communicate with each other through discussion forums.

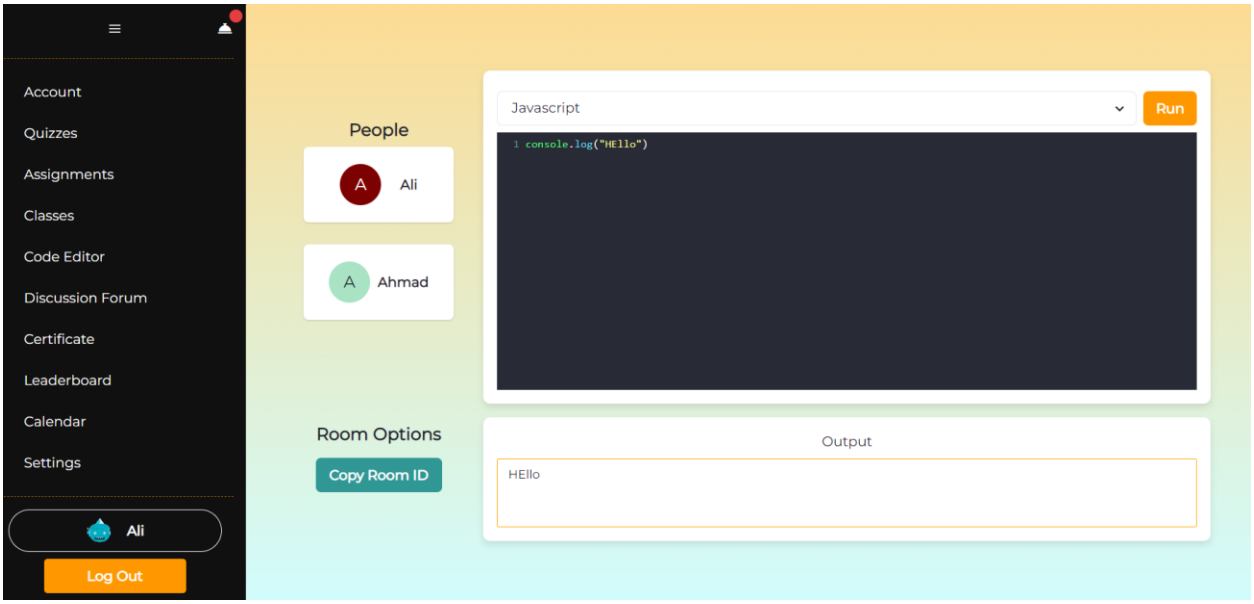


9.8 Collaborative Coding

9.8.1 Student's Point of View

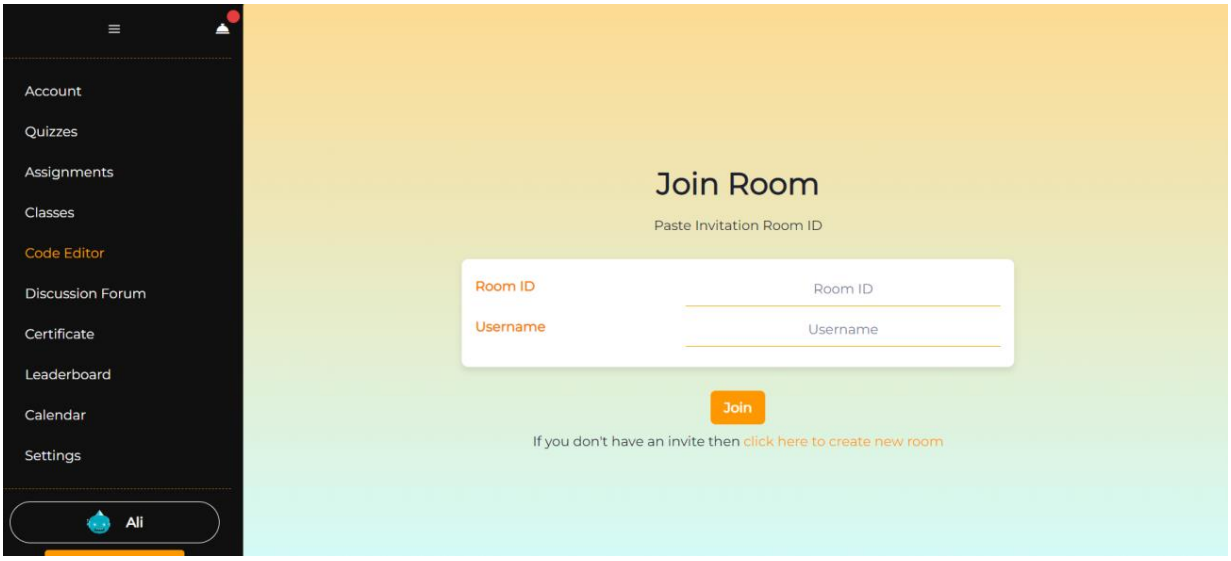
Students first generate the room ID by clicking on the "click here to create new room" and then copy the room ID, give it to the teacher that they want to see their code.

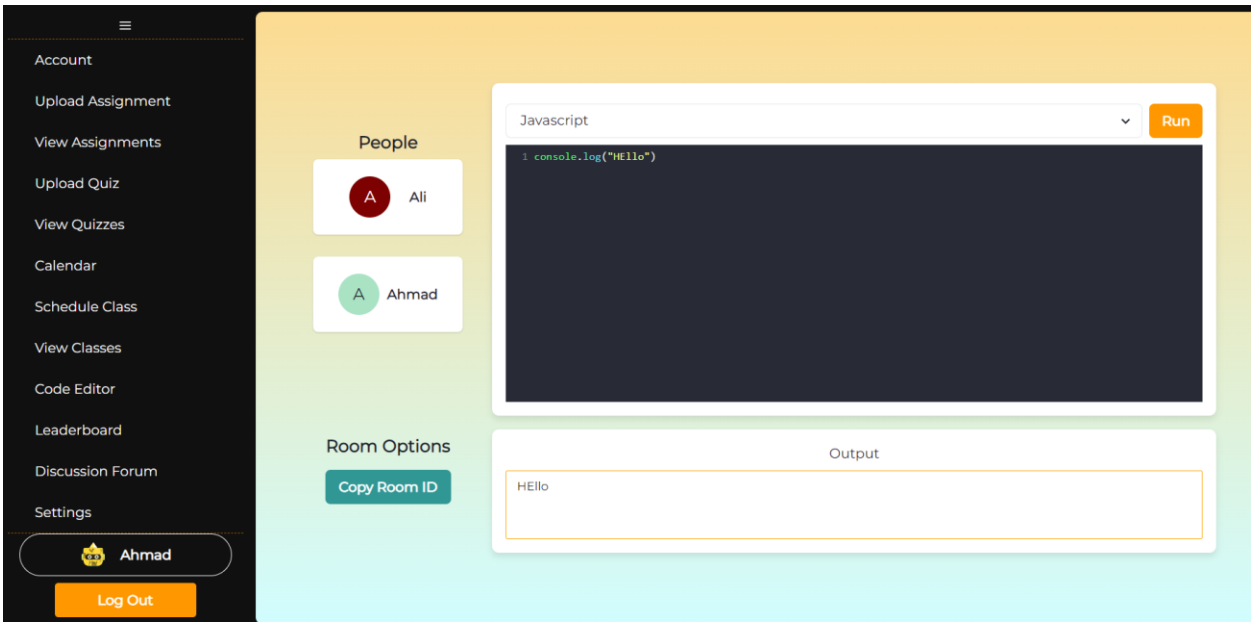




9.8.2 Teacher's Point of View

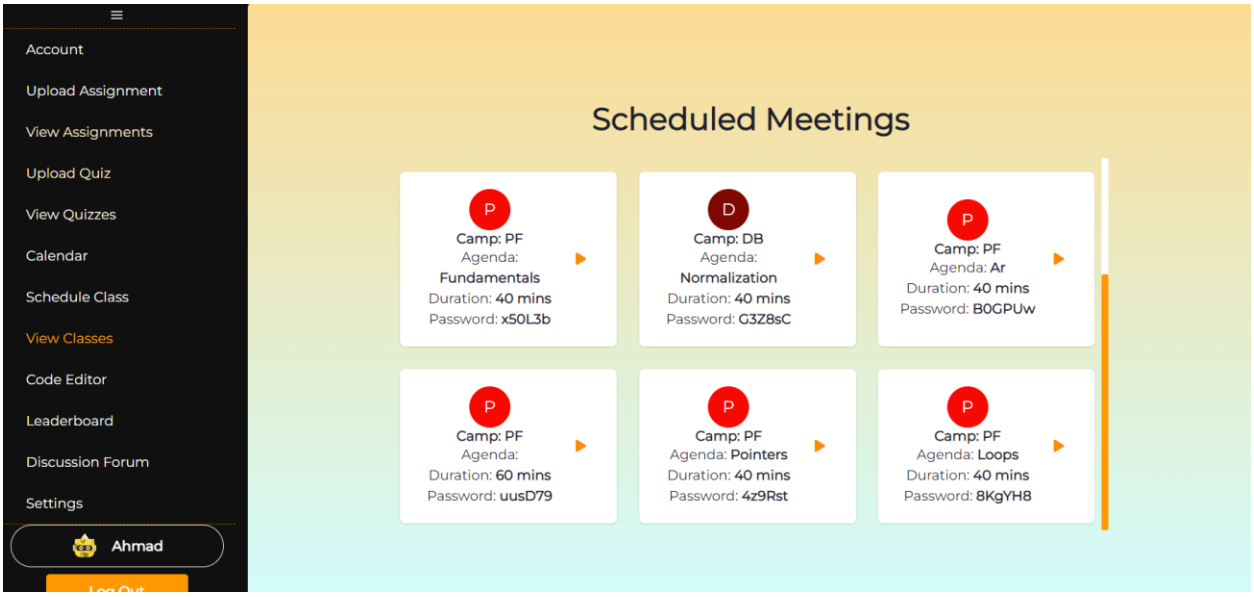
Teacher copy the room ID and paste it to join the room:

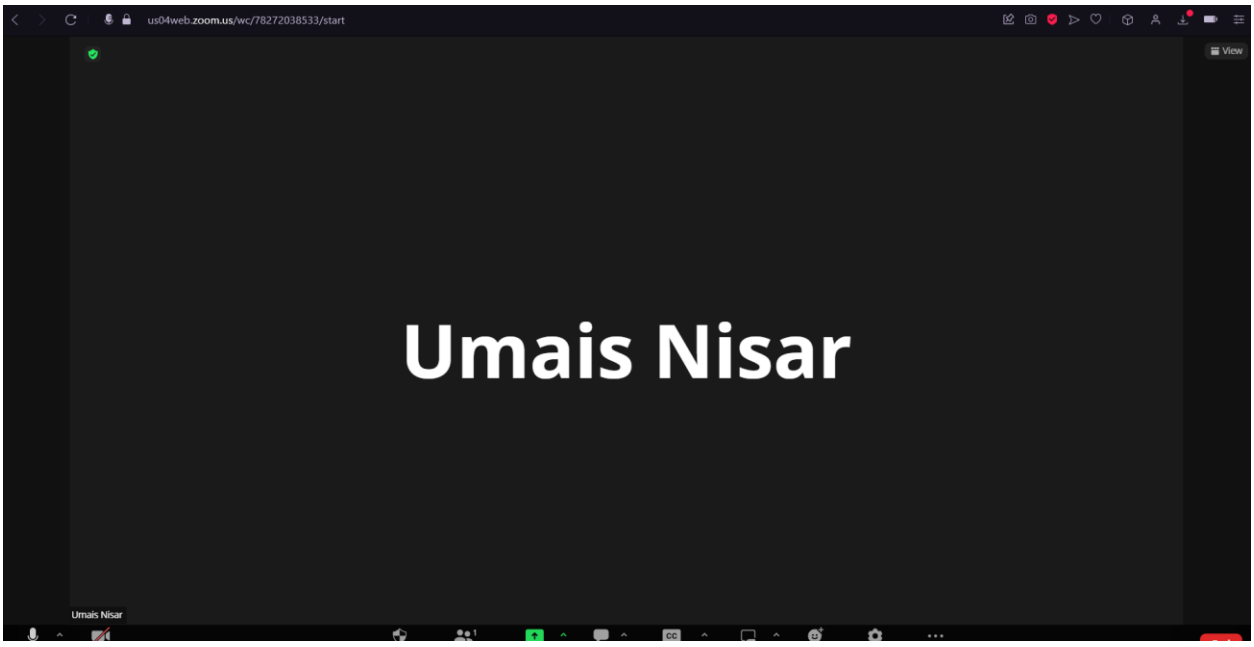




9.9 Online Classes

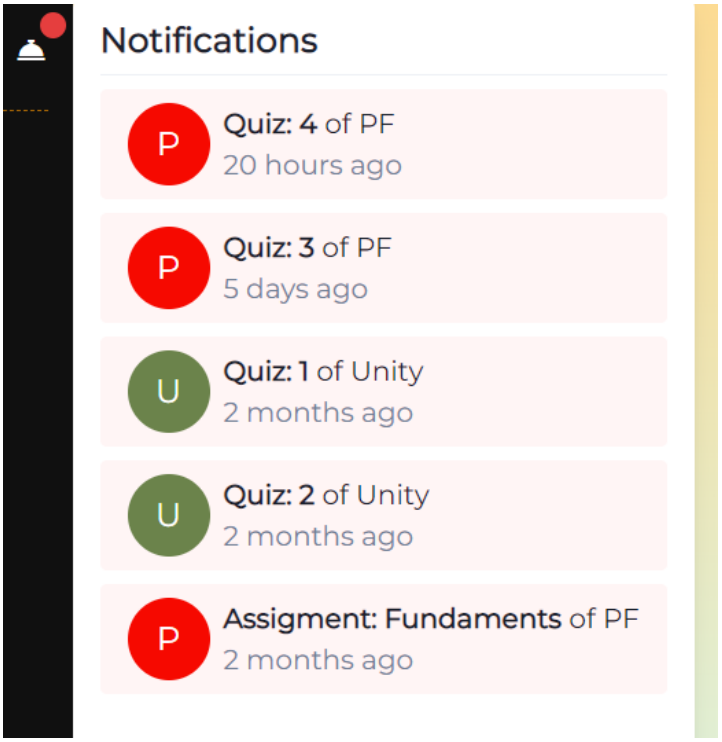
Teacher can create online classes and students can join them:





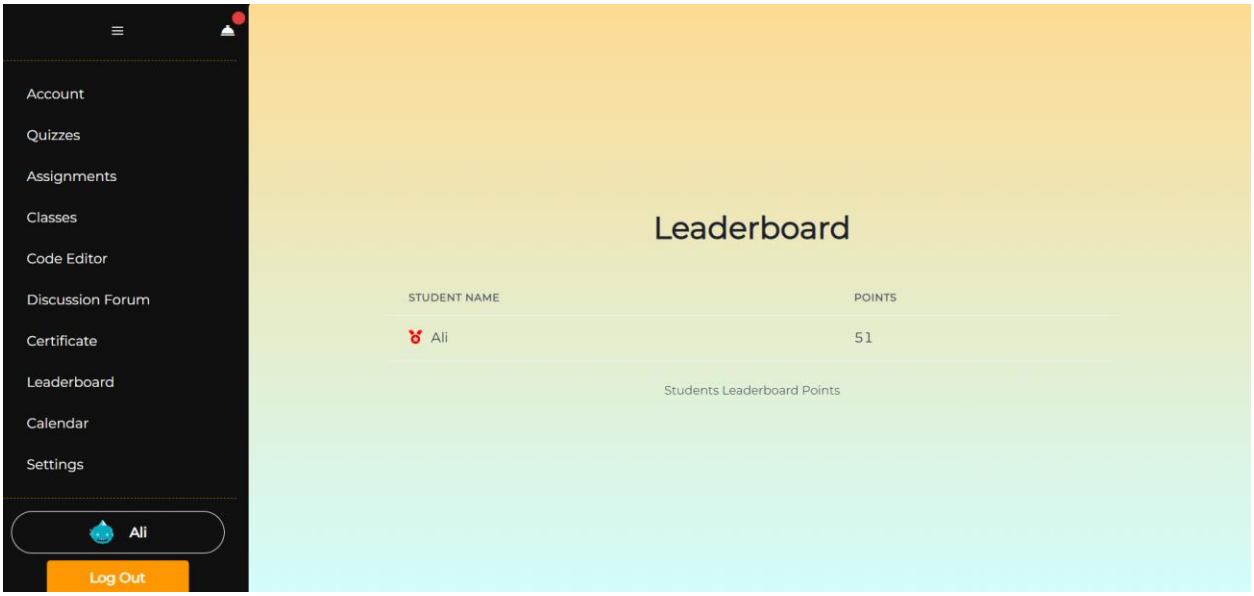
9.10 Notifications

Students are able to receive notifications for Assignments and Quizzes uploaded by the teacher:



9.10 Leaderboard

Both student and teacher can view progress of themselves and other students respectively:



Chapter 10: References

- *Judge0 CE - API Docs*, https://ce.judge0.com/. Accessed 3 June 2023.
- Carnes, Beau. “Learn the MERN Stack - Full Tutorial (MongoDB, Express, React, Node.js).” *YouTube*, 25 June 2019, https://www.youtube.com/watch?v=7CqJlxBYj-M. Accessed 23 October 2022.
- “Classes of E-Learning Management System Class Diagram.” *FreeProjectz*, 2 August 2017, https://www.freeprojectz.com/uml/e-learning-management-system-class-diagram. Accessed 23 October 2022.
- “CodeMirror Docs.” *CodeMirror*, https://codemirror.net/docs/. Accessed 3 June 2023.
- “Functional and Non-functional Requirements: Specification and Types.” *AltexSoft*, 23 July 2021, https://www.altexsoft.com/blog/business/functional-and-non-functional-requirements-specification-and-types/. Accessed 23 October 2022.
- “Introduction.” *Socket.IO*, 10 May 2023, https://socket.io/docs/v4/. Accessed 3 June 2023.
- “Learning resources - Material UI.” *MUI*, https://mui.com/material-ui/getting-started/learn/. Accessed 23 October 2022.
- Monsalve, Mauricio, and C. Larman. *Operation Contracts*, https://homepage.cs.uiowa.edu/~tinelli/classes/022/Spring15/Notes/chap11.pdf. Accessed 23 October 2022.
- “MVC Framework - Introduction.” *Tutorialspoint*, https://www.tutorialspoint.com/mvc_framework/mvc_framework_introduction.htm. Accessed 23 October 2022.

- “1 LAB 3: Introduction to Domain Modeling and Class Diagram OBJECTIVES Use the UML notation to represent classes and their pr.” *Nor Samsiah Sani's Weblog*, https://norsamsiah.files.wordpress.com/2010/01/lab-003-domain-modeling1.pdf. Accessed 23 October 2022.
- “Tutorial.” *WidgetBot*, https://docs.widgetbot.io/tutorial/. Accessed 3 June 2023.
- “UML - Class Diagram.” *Tutorialspoint*, https://www.tutorialspoint.com/uml/uml_class_diagram.htm. Accessed 23 October 2022.